

DESIGN AND IMPLEMENTATION OF A 2-DOF TWIN – ROTOR HELICOPTER TEST BED

A DESIGN REPORT

SUBMITTED FOR FINAL YEAR PROJECT

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FYP 05

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FEBRUARY, 2026

Summary

This project focuses on the design, modeling, and preliminary implementation of a 2-Degree-of-Freedom (2-DOF) twin rotor helicopter test bed intended for laboratory-scale control system experimentation. The system allows controlled motion in pitch and yaw axes, providing a practical platform to study nonlinear dynamics and control strategies relevant to aerospace and mechatronic applications.

The mechanical structure was designed using CAD tools, incorporating two A2212 1000 kV BLDC motors with 10×4.5 propellers, integrated ESCs, slip rings, and encoders to enable continuous rotation and real-time angle measurement. Structural materials such as 3D-printed ABS components and Aluminum 6061-T6 shafts were selected based on strength, weight, manufacturability, and fatigue considerations. Detailed stress, bending, torsion, and fatigue calculations were carried out to validate the structural safety of rotating components under worst-case loading conditions.

The system dynamics were formulated using the Newton–Euler approach, resulting in nonlinear equations of motion that describe pitch and yaw behavior. These equations were linearized around the equilibrium operating point to derive suitable transfer functions for controller design. A PID control strategy was selected as the initial control approach due to its simplicity, robustness, and suitability for real-time implementation.

For embedded control, an STM32 Cortex-F4 microcontroller was selected to execute control algorithms and interface with sensors and actuators. Preliminary plans include simulation of the system in MATLAB/Simulink, followed by real-time implementation and experimental validation of the control performance.

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Abbreviations

2 DOF	2 Degree of Freedom
MIMO	Multi Input Multi Output
SISO	Single Input Single Output
PID	Proportional-Integral-Derivative
3D	Three Dimensional
2D	Two Dimensional
POR	Point of Rotation

1. Introduction

The study of dynamic systems with more than one input and output is an important area in control engineering. Many practical systems used in aerospace, robotics, and mechatronics show multi-input multi-output (MIMO) behavior [1], [2], [3], [4] where the motion in one direction affects the motion in another. Understanding and controlling such coupled systems is essential for modern engineering applications. The two-degree-of-freedom (2-DOF) twin-rotor helicopter testbed is commonly used as a laboratory system to study these concepts in a safe and controlled environment.

This project involves the design and development of a 2-DOF twin-rotor helicopter testbed with coupled pitch and yaw motions. The system uses two motors to generate thrust, and each motor influences both the pitch and yaw angles. Due to this interaction, the system exhibits nonlinear and coupled MIMO dynamics similar to those found in real helicopter systems. This makes the testbed suitable for studying system behavior, modeling techniques, and control design.

In this stage of the project, a mathematical model of the system has been developed based on Newton and Euler's equations [2] to describe the nonlinear MIMO dynamics of the pitch and yaw motions and simplified for controller design. The nonlinear model is linearized around a chosen operating point. Through this process, the system is approximated as two single-input single-output (SISO) subsystems, one for pitch control and one for yaw control. These linear models provide a suitable basis for the design and simulation of PID controllers [5], [6].

Alongside the control system modeling, the mechanical design of the testbed has been carried out. The overall structure is divided into two main assemblies, a support structure, which provides stability and holds the fixed components, and a rotational structure, which allows controlled motion. The rotational structure is further divided into the pitch assembly and the yaw assembly to simplify design, analysis, and manufacturing.

3D models of the complete system have been created using SolidWorks. Based on these models, structural calculations have been performed to ensure that the components can safely withstand the expected loads during operation. Suitable materials have been selected by considering strength, weight, availability, and ease

of fabrication. In addition, detailed 2D engineering drawings have been prepared to support the manufacturing and assembly process.

This progress report presents the work completed so far, including mathematical modeling, mechanical design, structural analysis, material selection and power requirement analysis. These steps form the foundation for the next phase of the project, which will focus on PID controller design, simulation, hardware implementation, and experimental testing of the 2-DOF twin-rotor helicopter testbed.

2. Calculation and Material selection

2.1. Mathematical Model

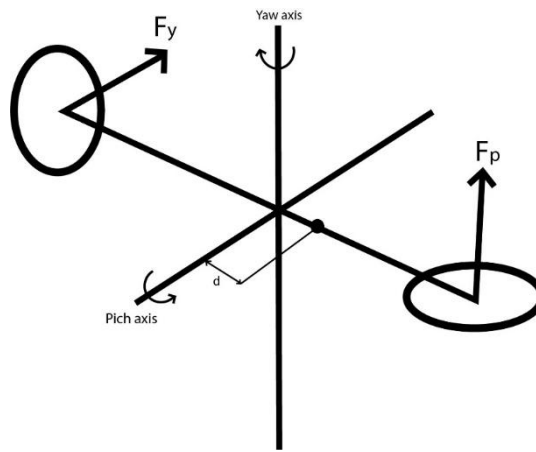


Figure 1 : Free body diagram

Pitch Angle = θ

Yaw Angle = φ

Euler's Rotational Equation,

$\sum \tau = \text{Sum of torques acting}$

$J = \text{moment of inertia}$

$\alpha = \text{Angular Acceleration}$

$$\sum \tau = J\alpha \quad \dots(1)$$

Defining J of equation (1),

From Parallel axis theorem, Let J_{cm} is inertia about center of mass, m is mass and d is distant between center of mass and point of rotation.

$$J_{point} = J_{cm} + md^2 \quad \dots(2)$$

For control designing we linearize around hover to approximating a nonlinear system by a linear one near a specific operating point.

Defining τ of equation (1),

Motor torque

Rotors generates thrust, and thrust at a distance produce torque, where F is the force generates by propellers and l is the distance to point of rotation.

$$\tau = Fl \quad \dots(3)$$

where T is thrust , torque constant define by K_f , ω^2 defines the angular acceleration.

Thrust from Rotor,

$$T = K_f \omega^2 \quad \dots(4)$$

ω^2 is nonlinear and for linear model ω depend on voltage so,

Introducing torque current relationship,

$$\tau_m = K_f I \quad \dots(5)$$

Where τ_m and I is motor torque and current. At steady state current 'I' can be written as,

$$I = \frac{v - K_e \omega}{R} \quad \dots(6)$$

So equation for motor torque can be rewritten as,

$$\tau_m = K_f \frac{v - K_e \omega}{R} \quad \dots(7)$$

This is still nonlinear so by linearizing around operating point,

$$\Delta \tau_m = KV \Delta V \quad \dots(8)$$

Where,

$$KV = \frac{K_v}{R} \quad \dots(9)$$

This give the control oriented model

$$\tau_m = K_f V \quad \dots(10)$$

K_{PP} , K_{YY} and V_p , V_y are pitch , yaw thrust coefficients and pitch , yaw motor voltages respectively,

$$\tau_m = K_{pp} V_p \text{ for pitch} \quad \dots(11)$$

$$\tau_m = K_{yy} V_y \text{ for yaw} \quad \dots(12)$$

Gravity torque

Gravity pulls the arm do wn,

$$\tau_g = mgd \sin\theta \quad \dots(13)$$

When linearized for small angles

$$\sin\theta \approx \theta$$

so equation becomes,

$$\tau_g = mgd \theta \quad \dots(14)$$

Damping torque

Damping resist motion

$$\tau_d = -D\alpha \quad \dots(15)$$

If θ , φ is pitch, Yaw angles then $\ddot{\theta}$, $\ddot{\varphi}$ is pitch, yaw acceleration respectively and D_p , D_y is viscous damping coefficient for pitch and yaw respectively.

$$\tau_d = -D_p \ddot{\theta} \text{ for pitch} \quad \dots(16)$$

$$\tau_d = -D_p \ddot{\varphi} \text{ for yaw} \quad \dots(17)$$

Then equation (1) can be rewritten individually for pitch and yaw as,

For Pitch,

$$J_p \ddot{\theta} = K_{pp} V_p - D_p \ddot{\theta} - mgd \theta \quad \dots(18)$$

Dividing both side by J_p above equation can be written as,

$$\ddot{\theta} + \frac{D_p}{J_p} \dot{\theta} + \frac{mgd}{J_p} \theta = \frac{K_{pp}}{J_p} V_p \quad \dots(19)$$

For Yaw,

Yaw is Horizontal – No gravitational Torque,

$$J_y \ddot{\varphi} = K_{yy} V_y - D_y \ddot{\varphi} \quad \dots(20)$$

Dividing both side by J_y above equation can be written as,

$$\ddot{\varphi} + \frac{D_y}{J_y} \dot{\varphi} = \frac{K_{yy}}{J_y} V_y \quad \dots(21)$$

2.2. Materials Selection

2.2.1. 3D Printing Material Selection

1. 3D printing parts
 - Support structure
 - Propellers case
 - yaw structure
 - Pitch structure
 - Housing
2. 3D printing materials,
 - ABS
 - PETG
 - Nylon
 - PLA
3. Our requirements to select 3D printing material
 - Reduction in weight
 - High strength to weight ratio in the parts mainly the rotor blades and structural frame must be as light possible to ensure efficient performance but strong enough to withstand operational force.
 - Tensile strength & flexural strength the material needs to resist breaking under tension and bending.
 - Fatigue resistance is needed because, as a test bed the components will be subjected to repeated load / unload cycles (vibrations & rotor rotation). The material must be developing cracks or fail over time due to dynamic stress.
 - Thermal stability & Environmental resistance.
 - Good layer adhesion
 - Printability & cost effectiveness
 - Post processing
 - Availability
 - Durability in dynamic systems
 - Surface finish
 - Dimensional stability
 - Mechanical strength

In the above requirement we decided to use ABS material for main frame and PETG for the down part covering.

Table 1 : Comparison between ABS and PETG

Property	ABS	PETG
Mechanical strength	High impact resistance and good toughness suitable for structural and bearing components (1.75 mm)	Good strength & toughness but slightly more flexible than ABS reducing risk of brittle failure. (1.75)
Thermal resistance	High heat resistance (glass transition temp=105°C) making it suitable for parts exposed to moderate heat.	Lower heat resistance (glass transition temp=80°C.) less suitable for high temperature environment.
Printability	More difficult to print require a heat bed and enclosed chamber to minimize wrapping	Easier to print minimal wrapping does not strictly require an enclosure.
Dimensional stability	Prone to warping and shrinkage if printing conditions are not well controlled	Excellent dimensional stability with low shrinkage.
Surface finish	Matte finish	Naturally glossy surface
Post processing	Easy to sand, drill & chemically smooth	It is easy to drill and sand but limited chemical post processing.
Durability in dynamic system	Well suited for dynamic and vibration prone components.	Good durability but flexibility may reduce stiffness in precision parts.
Cost & Availability	Widely available and cost effective	It is slightly more expensive than ABS but still affordable.

In the context of the 2-DOF twin rotor helicopter testbed, ABS is more suitable for structural and load bearing components such as motor mounts, bearing housing and mechanical arms due to its higher thermal resistance and rigidity.

These properties are essential for maintaining dimensional accuracy and mechanical stability under continuous motor operation & vibration.

2.2.2. Shaft Material Selection

Table 2 : Candidate Aluminum Alloys Considered

Alloy	Typical Use	Key Feature
Al 2024-T3	Aircraft structures	High strength, poor corrosion
Al 7075-T6	Aerospace & defence	Very high strength
Al 6061-T6	Structural & mechanical parts	Balanced properties

Table 3 : Key Material Properties Comparison

Property	Al 2024-T3	Al 7075-T6	Al 6061-T6
Density (kg/m ³)	2780	2810	2700
Yield strength (MPa)	~325	~503	276
Ultimate strength (MPa)	~470	~572	310
Fatigue resistance	Moderate	Moderate	Good
Corrosion resistance	Poor	Poor	Good
Machinability	Fair	Difficult	Excellent
Weld ability	Poor	Poor	Excellent
Cost	Medium	High	Low
Availability	Limited	Limited	High

Why Aluminum 6061-T6 is Selected?

1. Sufficient Strength for the Project

- Structural and shaft stresses in the testbed are **far below 276 MPa**.
- Higher-strength alloys (7075-T6) are **not required**, making them unnecessary over-design.

2. Lower Weight Improves Control Performance

- 6061-T6 has the **lowest density** among the candidates.
- Lower mass reduces Rotational inertia, Required thrust and Motor load.
- This directly improves **pitch and yaw response**.

3. Better Fatigue Performance for Rotating Components

- The testbed operates under **repeated cyclic loading**.
- 6061-T6 offers **more stable fatigue behavior** compared to 2024-T3 and 7075-T6, which are more notch-sensitive.

4. Excellent Machinability & Fabrication

- Easy to machine using standard workshop tools.
- Allows precise shaft, bearing seat, and coupling fabrication.
- Ideal for academic laboratories where frequent modifications are expected.

5. Good Corrosion Resistance

- Naturally forms a protective oxide layer.
- No need for surface treatments in indoor lab environments.
- 2024-T3 and 7075-T6 require protective coatings.

6. Weld ability and Assembly Advantages

- 6061-T6 can be welded or joined easily.
- Other alloys suffer severe strength loss after welding.

7. Cost and Availability

- 6061-T6 is **widely available** and **cost-effective**.
- Suitable for student projects with limited budgets.

Why the Other Alloys Were NOT Selected

- Aluminum 2024-T3
 - Poor corrosion resistance
 - Difficult to weld
 - Overqualified strength for this application
- Aluminum 7075-T6
 - Excessively high strength (not required)
 - Brittle behavior under fatigue
 - Difficult machining
 - High cost

2.3. Calculations

2.3.1. Structure Calculation

2.3.1.1. ABS structural Thickness Determination for 3D Printed Rotating components

The primary rotating structural Component (main axis/arm) has the following geometry.

Dimensions of Pitch Structure

- Length (L) = 30 cm = 0.3 m
- Width (b) = 4 cm = 0.04 m
- Height = 3 cm = 0.03 m
- Distance from POR to Motor = 28.5cm = 0.285m
- Thickness (h) = to be determined

Loading Conditions

- Maximum Thrust produced by one motor = 800 g = 0.8 kg
- Gravitational acceleration = 9.81 m/s²
- Applied force $F = 0.8 \times 9.81 = 7.85$ N

Worst -case Loading is assumed as cantilever bending. Which produces the highest stress.

Material Properties of ABS

Yield Tensile Strength	34 MPa
Ultimate Tensile Strength	41 MPa
Tensile Modulus	2 GPa
Shear Strength	67 MPa
Shear Modulus	2 GPa
Fatigue Strength	17 MPa
Elongation At Break	30 %
Poisson's Ratio	0.35
Hardness	H358/30
Density	1.1 g/cm ³
Impact Strength (ASTM)	347 J/m

Figure 2 : Mechanical Properties of ABS

From Figure 2,

- Ultimate tensile strength:

$$\sigma_{UTS} = 41 \text{ MPa}$$

- Safety factor: (20%) Additional
FOS = 1.2

- Allowable stress:

$$\sigma_{allow} = \frac{\sigma_{UTS}}{FOS} = \frac{41}{1.2} = 34.17 MPa \quad \dots(22)$$

Bending stress calculation (Governing Stress)

- Bending moment

$$M = F L \quad \dots(23)$$

$$M = 7.85 \times 0.285 = 2.23725 \text{ Nm}$$

- Bending Stress Equation (Rectangular Section)

$$\sigma_b = \frac{6M}{bh^2} \quad \dots(24)$$

- Minimum Thickness Determination

Rearranging for h:

$$h = \sqrt{\frac{6M}{b\sigma_{allow}}} \quad \dots(25)$$

- Substituting values:

$$h = \sqrt{\frac{6 \times 2.23725}{0.04 \times 34.17 \times 10^6}} = \underline{\underline{3.1 \text{ mm}}}$$

Selected Thickness

h selected = 5 mm

- meets the requirement $< 3.1 \text{ mm}$
- Provides additional stiffness and
- Vibration resistance

Bending Stress verification (5 mm Thickness)

- Second Moment of Area.

$$I = \frac{bh^3}{12} \quad \dots(26)$$

$$I = \frac{0.04 \times (0.005)^3}{12} = 4 \times 10^{-9} m^4$$

- Maximum Bending Stress.

$$\sigma_b = \frac{My}{I} \quad \dots(27)$$

$$y = \frac{h}{2} = 0.0025m$$

$$\sigma_b = \frac{2.23725 \times 0.0025}{4 \times 10^{-9}} = 14 MPa$$

$$\sigma_b < \sigma_{allow} \text{ (SAFE)}$$

Torsion Stress calculation.

Yaw motion produces torsion in the rotating structure.

- Torque (T) = 2.23725 Nm
- Approximate polar Moment of Area

$$J \approx \frac{bh^3}{3} \quad \dots(28)$$

$$J = \frac{0.04 \times (0.005)^3}{3} = 1.67 \times 10^{-9} m^4$$

- Maximum Torsional Stress

$$\tau = \frac{Tr}{J} \quad \dots(29)$$

$$r = 0.0025m$$

$$\tau = \frac{2.23725 \times 0.0025}{1.67 \times 10^{-9}}$$

$$\tau = 3.35 \text{ MPa (well below. ABS Shear Capacity.)}$$

Shear Stress Calculation.

- Cross-sectional Area.

$$A = b \times h = 0.04 \times 0.005 = 2 \times 10^{-4} m^2$$

- Average shear stress,

$$\tau = \frac{F}{A} \quad \dots(30)$$

$$\tau = \frac{7.85}{2 \times 10^{-4}} = 0.03925 \text{ MPa (Negligible compared to ABS shear strength)}$$

Fatigue stress consideration

- Loading is cycle due to
- Continuous motor Operation
- Pitch and yaw oscillations
- Propeller -induced vibration
- Maximum Stress amplitude: $\sigma_{\max} = 2.83 \text{ MPa}$
- Stress level is <10% of ABS tensile Strength,
- ABS has good toughness and damping, making it suitable for cyclic loading fatigue failure is not expected during lab operation.

2.3.1.2. Aluminum 6061-T6

Physical Properties	Metric	English	Comments
Density	2.7 g/cc	0.0975 lb/in ³	AA; Typical
Mechanical Properties			
Hardness, Brinell	95	95	AA; Typical; 500 g load; 10 mm ball
Hardness, Knoop	120	120	Converted from Brinell Hardness Value
Hardness, Rockwell A	40	40	Converted from Brinell Hardness Value
Hardness, Rockwell B	60	60	Converted from Brinell Hardness Value
Hardness, Vickers	107	107	Converted from Brinell Hardness Value
Ultimate Tensile Strength	310 MPa	45000 psi	AA; Typical
Tensile Yield Strength	276 MPa	40000 psi	AA; Typical
Elongation at Break	12 %	12 %	AA; Typical; 1/16 in. (1.6 mm) Thickness
Elongation at Break	17 %	17 %	AA; Typical; 1/2 in. (12.7 mm) Diameter
Modulus of Elasticity	68.9 GPa	10000 ksi	AA; Typical; Average of tension and compression. Compression modulus is about 2% greater than tensile modulus.
Notched Tensile Strength	324 MPa	47000 psi	2.5 cm width x 0.16 cm thick side-notched specimen, K _t = 17.
Ultimate Bearing Strength	607 MPa	88000 psi	Edge distance/pin diameter = 2.0
Bearing Yield Strength	386 MPa	56000 psi	Edge distance/pin diameter = 2.0
Poisson's Ratio	0.33	0.33	Estimated from trends in similar Al alloys.
Fatigue Strength	96.5 MPa	14000 psi	AA; 500,000,000 cycles completely reversed stress; RR Moore machine/specimen
Fracture Toughness	29 MPa-m ^{1/2}	26.4 ksi-in ^{1/2}	K _{IC} ; TL orientation.
Machinability	50 %	50 %	0-100 Scale of Aluminum Alloys
Shear Modulus	26 GPa	3770 ksi	Estimated from similar Al alloys.
Shear Strength	207 MPa	30000 psi	AA; Typical

Figure 3 : Mechanical Properties of Aluminum 6061-T6

Why select that material

- (i) Light weight
- (ii) Good fatigue resistance
- (iii) Commonly used for rotating Shafts
- (iv) Easy to machine

From Figure 3,

Yield Strength

$$\sigma_y = 276 \text{ MPa}_{(\text{max})}$$

Ultimate tensile Strength

$$\sigma_{UTS} = 310 \text{ MPa}_{(\text{max})}$$

Shear yield strength (ductile material)

$$\tau_y = 0.58\sigma_y$$

$$\sigma_y = 160 \text{ MPa}_{(\text{max})}$$

- Safety Factor - FOS -1.2 (20% Additional)
- Allowable Stress,

$$\tau_{allow} = \frac{160}{1.2} = 133 \text{ MPa}$$
- Allowable shear Stress = 133 MPa
- Allowable bending stress $\sigma_{allow} = \frac{276}{1.2} = 230 \text{ MPa}$

Minimum for Allowable Diameter Calculation Aluminum Rotating shaft

Given Design Data,

- Thrust force

$$F = mg \quad \dots(31)$$

$$\begin{aligned} F &= 0.8 \times 9.81 \\ &= 7.85 \text{ N} \end{aligned}$$

- Distance from bearing to motor 2
285 mm = 0.285 m
- Shaft Subjected to
 - (i) Torsion (due to thrust)
 - (ii) Bending (due to offset load)
 - (iii) Cyclic loading (fatigue)

- Load Calculations.
Torque on a shaft,

$$T = FL \quad \dots(32)$$

$$\begin{aligned} T &= 7.85 \times 0.285 \\ T &= 2.237 \text{ Nm} \end{aligned}$$

- Bending Moment
 $M = FL$
 $M = 2.237 \text{ Nm}$

Minimum Diameter Determination

The shaft is under the Combined torsion and bending, so the following are the Calculations of Minimum diameter from both Criteria & Select the governing value.

governing value = the governing value is the most critical value that controls the final design decision

- Minimum Diameter from Torsion
Minimum outer diameter of Hollow Shaft.

Outer diameter = D

inner diameter = d

- Define diameter ratio.

$$k = \frac{d}{D} \quad k - \text{diameter ratio}$$

- A Common design assumption
K = 0.6 (weight efficient & realistic)
- Torsional stress Equation (Hollow shaft)

$$\tau = \frac{16T}{\pi D^3(1-k^4)}$$

τ - Torsional Stress T - Applied torque D³ - Diameter

- for safe design
 $\tau \leq \tau_{allow}$
- Rearranging to find minimum outer diameter

$$D_{min} = \left(\frac{16T}{\pi \tau_{allow} (1 - k^4)} \right)^{1/3} \quad \dots(33)$$

$$= \left(\frac{16 \times 2.237}{\pi \times 133 \times 10^6 \times (1 - 0.6^4)} \right)^{1/3}$$

$$= 1 - 0.6^4$$

$$= 0.8704$$

$$D_{min} = 4.7 \text{ mm}$$

- Bending stress for hollow shaft,

$$\sigma_b = \left(\frac{16T32M}{\pi D^3 (1 - k^4)} \right)^{1/3} \quad \dots(34)$$

$$D_{min, \text{ bending}} = 4.2 \text{ mm}$$

- Governing Diameter,
 $D_{\min} = \max(4.7, 4.2) = 4.7 \text{ mm}$
- Minimum wall thickness of hollow shaft,

$$d = kD \quad \dots(35)$$

$$t = \frac{D - d}{2} = \frac{D(1 - k)}{2} \quad \dots(36)$$

$$D = 4.72 \text{ mm} \quad k = 0.6$$

$$t_{\min} = \frac{4.72(1-0.6)}{2} = 0.94 \text{ mm}$$

$$t_{\min} \approx 1 \text{ mm}$$

- Fatigue verification

Endurance limit

$$\sigma_e \approx 0.45\sigma_y$$

$$\sigma_e = 0.45 \times 276 = 124 \text{ MPa}$$

- Equivalent Stress

$$\sigma_{eq} \sqrt{\sigma_b^2 + 3\tau^2} \quad \dots(37)$$

From previous calculation:

$$\sigma_b = 2.83 \text{ MPa} \quad \tau = 0.71 \text{ MPa}$$

$$\sigma_{eq} = \sqrt{(2.83)^2 + 3(0.71)^2}$$

$$\sigma_{eq} = \sqrt{8.01 + 1.51}$$

$$\sigma_{eq} = 3.09 \text{ MPa}$$

$$\sigma_{eq} \ll \sigma_e$$

$$3.09 \text{ MPa} \ll 124 \text{ MPa}$$

Final governing values

Minimum outer diameter 4 - 7 mm

Minimum wall thickness 1.033

Governing failure mode Torsion

Material Al 6061-T6

Safety factor 1.2

2.3.2. Hardware Calculation

2.3.2.1. Motor - A2212 | 13 T | 1000kv | BLDC Motor

Dr Kiwi test results									
Propeller	Gear Ratio	Volts	Amps	Watts	RPM	Speed (mph)	Thrust (g)	Thrust (oz)	RPM as % of Kv*V
GWS HD 8x4	1	7	3.35	23	6630	25.1	226	7.97	88%
GWS HD 8x4	1	7.9	4.1	32	7410	28.1	287	10.12	87%
GWS HD 8x4	1	8.9	4.85	43	8220	31.1	347	12.24	86%
GWS HD 8x4	1	9.9	5.65	55	8940	33.9	420	14.82	84%
GWS HD 8x4	1	10.9	6.5	70	9660	36.6	495	17.46	82%
GWS HD 9x5	1	6.9	5.5	37	6000	28.4	348	12.28	81%
GWS HD 9x5	1	7.9	6.7	52	6660	31.5	436	15.38	78%
GWS HD 9x5	1	8.9	7.85	69	7290	34.5	526	18.55	76%
GWS HD 9x5	1	9.9	9.25	91	7920	37.5	627	22.12	74%
APC E 10x5	1	6.9	7	48	5610	26.6	406	14.32	75%
APC E 10x5	1	7.9	8.45	66	6120	29.0	505	17.81	72%
APC E 10x5	1	8.9	9.9	88	6690	31.7	604	21.31	70%
APC E 10x5	1	9.9	11.45	113	7170	34.0	702	24.76	67%
APC E 10x5	1	10.9	13	141	7650	36.2	802	28.29	65%
GWS HD 10x6	1	6.9	7.2	49	5610	31.9	424	14.96	75%
GWS HD 10x6	1	7.9	8.7	68	6180	35.1	526	18.55	72%
GWS HD 10x6	1	8.9	10.1	89	6690	38.0	617	21.76	70%
GWS HD 10x6	1	9.9	11.7	115	7200	40.9	722	25.47	67%
GWS HD 10x6	1	10.9	13.25	144	7680	43.6	817	28.82	65%
GWS HD 10x8	1	10.8	18.2	196	6390	48.4	733	25.86	55%

Figure 4 : Thrust Analysis of A2212 1000kv Motor

- Total lift force required per motor,
 $M = 0.800 \text{ kg}$ (Max thrust)

$$g = 9.81 \text{ m/s}^2$$

$$F = Mg$$

$$= 0.8 \times 9.81 = 7.848 \text{ N}$$

- Thrust required per motor,
 $1 \text{ N} = 101.97 \text{ g}$

$$7.848 \times 101.97 = \underline{\underline{800.26056 \text{ g}}}$$

- Motor speed calculation,
 $N = KV \times V$

$$N - \text{Motor speed (RPM)} \quad K_v - \text{Motor speed constant (RPM/V)}$$

$$V - \text{Supply voltage (V)}$$

Where,

$$KV = 1000 \text{ RPM / V}$$

$$V = 11.1 \text{ V}$$

$$N = 1000 \times 11.1 = 11100 \text{ RPM (No - Load)}$$

$$N = K_v \times V$$

$$N_{\text{Loaded}} = 65\% \text{ to } 75\% \text{ of no-load speed,}$$

$$N_{\text{loaded}} \approx 0.7 \times 11100 = 7770 \text{ RPM}$$

- Angular Velocity,

ω - Angular velocity (rad/s) N - Rotational speed (RPM)

$$\omega = \frac{2\pi N}{60}$$

$$\omega = \frac{2\pi \times 7770}{60} \approx 813.6 \text{ rad/s}$$

- Torque Calculation,

Assuming motor power using motor data sheet,

$$P = 150 \text{ W}$$

Assume efficiency $\approx 80\%$

Then,

$$P_{\text{mech}} = 0.8 \times 150 = 120 \text{ W}$$

$$\tau = \frac{P}{\omega}$$

τ - Torque (Nm)

P - Mechanical output power (W)

ω - Angular velocity (rad/s)

$$\tau = \frac{120}{813.6} = 0.147 \text{ Nm}$$

2.3.2.2. ESC current rating

- Maximum Motor current,

Assuming motor current using motor data sheet,

$$I_{\text{motor}} = 13\text{A}$$

So then Applying Safety Fact

I_{ESC} - Required ESC current rating Ampere (A)

$$I_{\text{ESC}} = \beta \times I_{\text{motor}}$$

I_{motor} - Maximum motor current Ampere (A)

β - Safety factor (1.2-1.3) dimensionless

$$I_{\text{ESC}} = 1.3 \times I_{\text{motor}}$$

$$I_{\text{ESC}} = 1.3 \times 13 = 16.9\text{A}$$

So, then we selected ESC – 20A or 30A CW & CCW

2.3.3 Power Requirement Analysis

Power Supply

Table 4 : Current Rating for all Electrical Components

Component	Voltage (V)	Ampere (A)
Two Motor current usage (Safety factor Added)	11.1	2×15.6
STM 32 Microcontroller	12	0.1
Two Encoders	12	0.16
Total	12	2×15.78

But We decided to apply Multiple channel output power supply. (Like this PSU - CP E SNT 250W 12V 21A | WEIDMULLER) So then,

Table 5 : Current Ratings After Considering Multiple Channel Power Supply

Component	Voltage (V)	Ampere (A)
Motor current usage (Safety factor Added)	11.1	15.6
STM 32 Microcontroller	12	0.1
Encoders	12	0.08
Total	12	15.78

With margin = 20A

- So then, Power rating,

$$P = V * I$$

$$P = 12 * 20$$

$$= 240w$$

3. Engineering Drawings

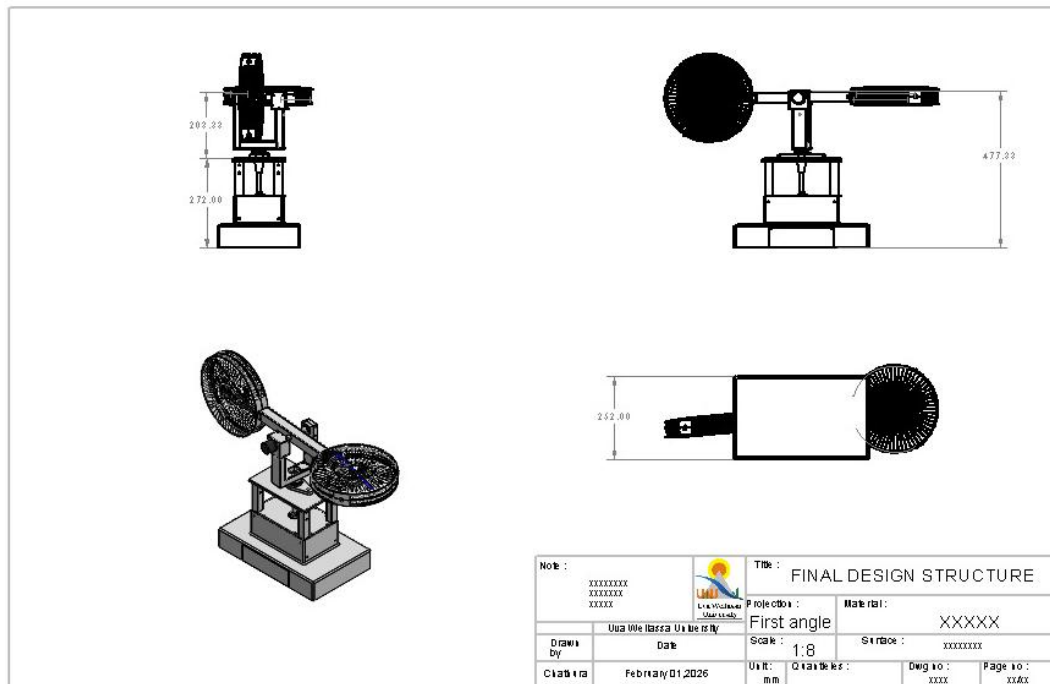


Figure 5 : 2D Drawing of Final Design Assembly

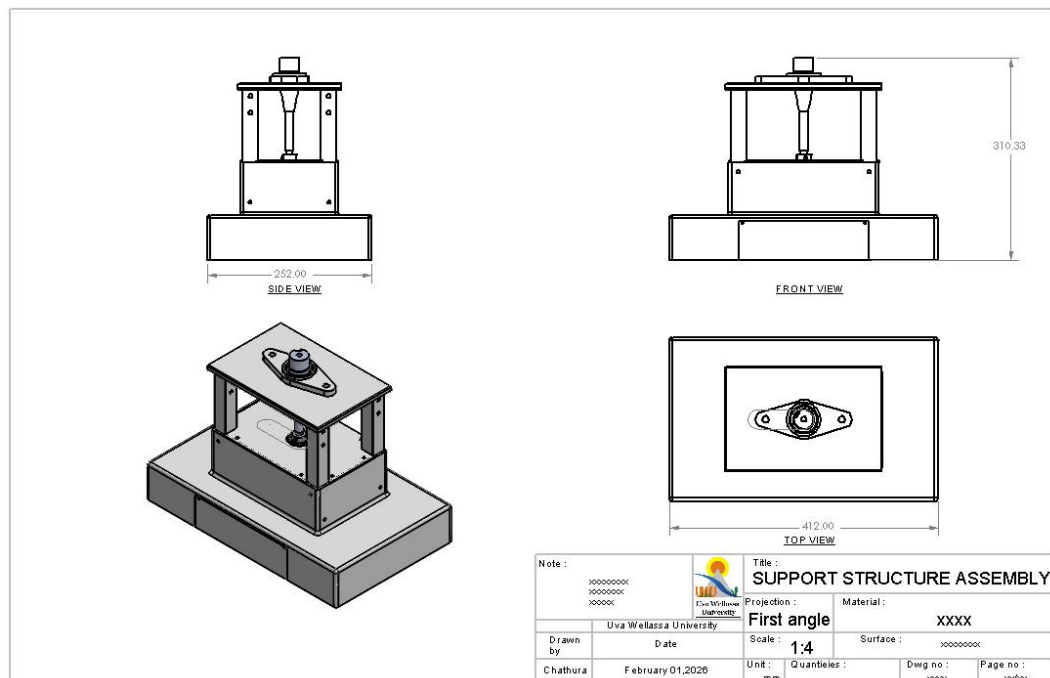


Figure 6 : 2D Drawing of Support Structure Assembly

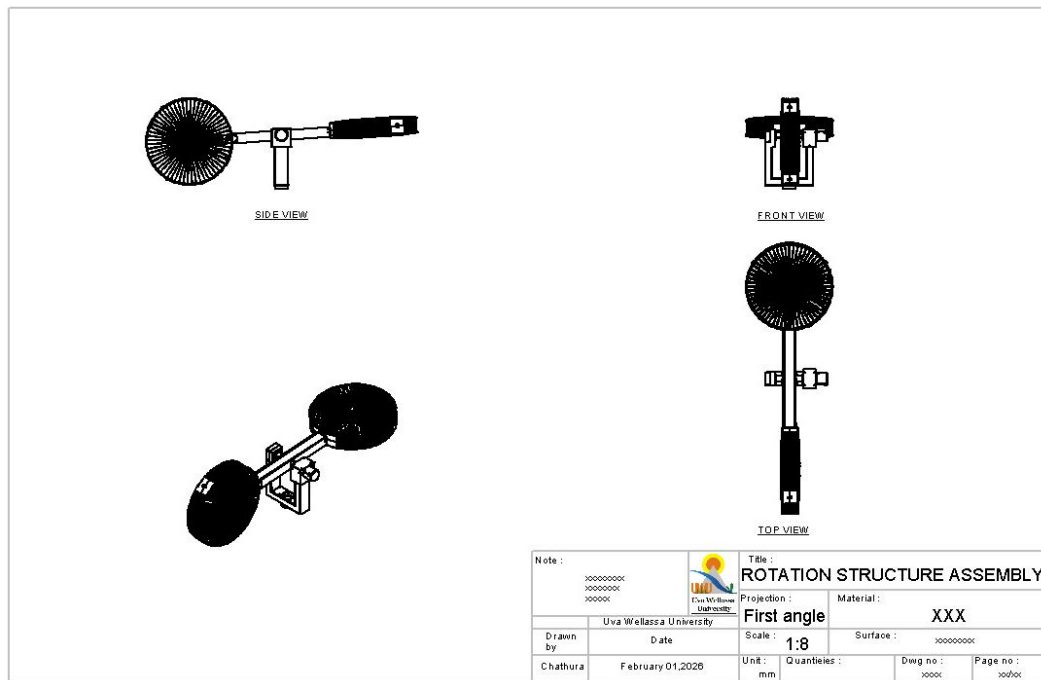


Figure 7 : 2D Drawing of Rotation Structure Assembly

All Individual Components Drawings are attached in appendices.

4. Analysis section

A symmetric 2-DOF helicopter structure was designed using SolidWorks to allow free rotational motion in both pitch and yaw axes. The structure consists of two main rotating subsystems: the pitch structure and the yaw structure, and support Structure.

4.1. Rotation Structure

Pitch Structure:

The pitch structure consists of:

- Two A2212 / 1000 KV BLDC motors
- Two 10×4.5 (1045) propellers
- Two propeller cases
- A rigid horizontal pitch beam

The two BLDC motors are mounted symmetrically at the ends of the pitch beam to generate lift and control torque. The ESCs are integrated inside the main pitch structure, which improves cable management, protects electronics, and maintains a clean mechanical layout.

Yaw Structure:

The yaw structure consists of:

- An outer box frame where all wiring arranged inside
- An MD0846 rotary encoder for yaw angle measurement

4.2. Support Structure

- A rigid support structure that houses:
 - A second rotary encoder
 - The STM32 Cortex-F4 microcontroller
 - The power supply unit

A **timing belt and pulley mechanism** is used to transmit the yaw rotational motion from the main vertical shaft to the encoder shaft. This arrangement allows accurate yaw angle sensing while isolating the encoder from mechanical shock and misalignment.

The overall mechanical design was optimized for:

- Structural rigidity
- Symmetric mass distribution
- Smooth pitch and yaw rotation
- Safe integration of electronics
- Ease of assembly and maintenance

4.3. Force Simulation of Pitch Structure

Only pitch motor load directly affects pitch structure so, maximum thrust force generated from pitch motor 7.85N. But for simulation, lets assuming safe continues thrust force is 7N,

Simulation Results

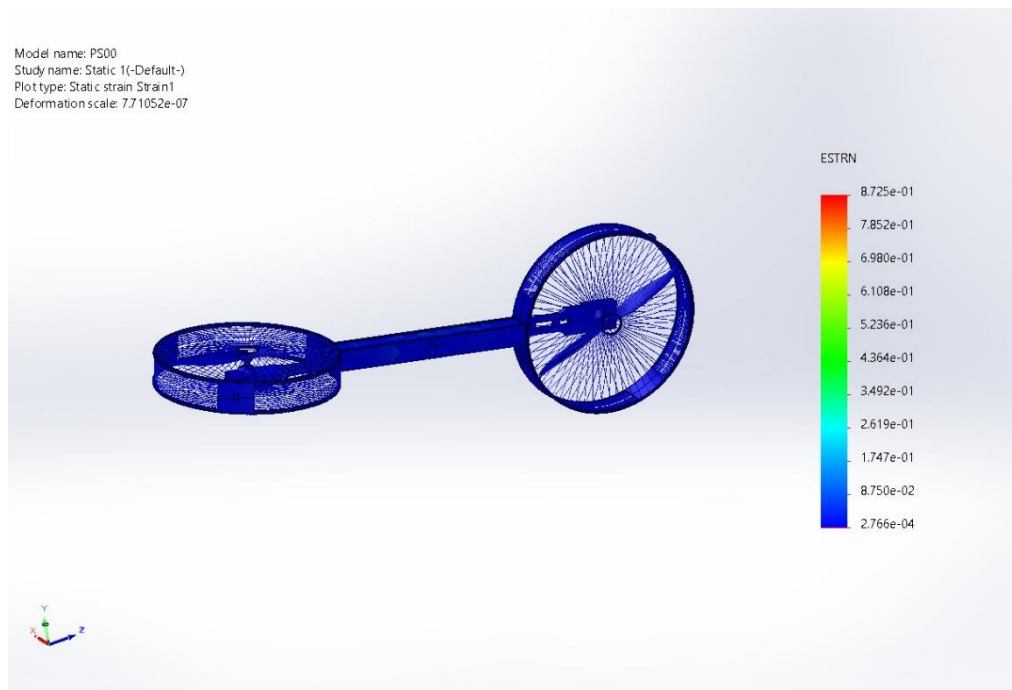


Figure 8 : Strain on Pitch Structure

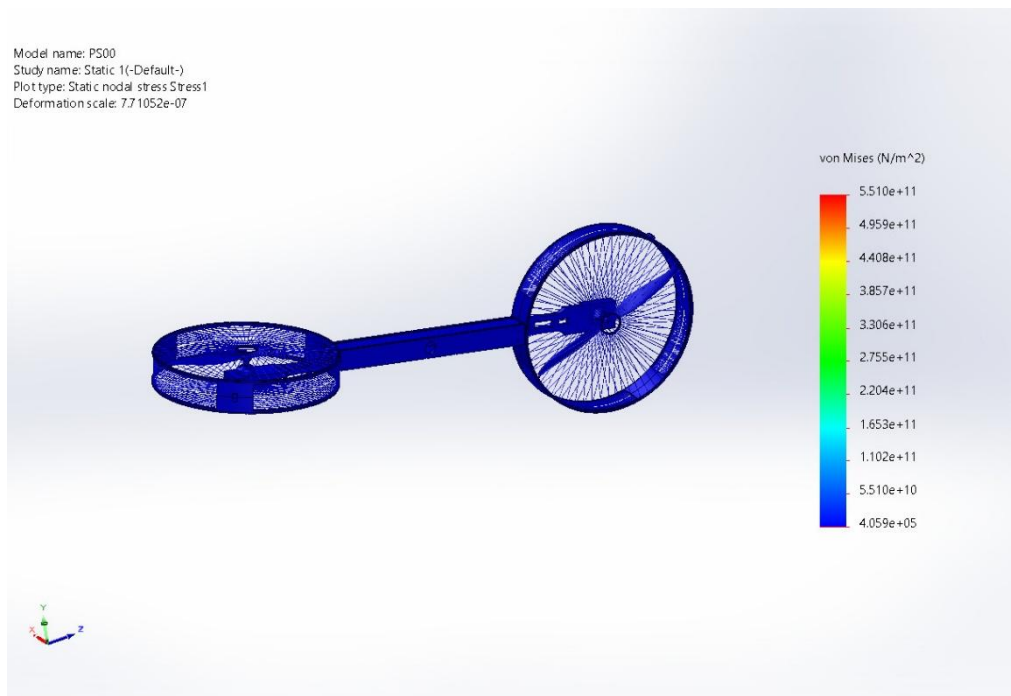


Figure 9 : Stress on Pitch Structure

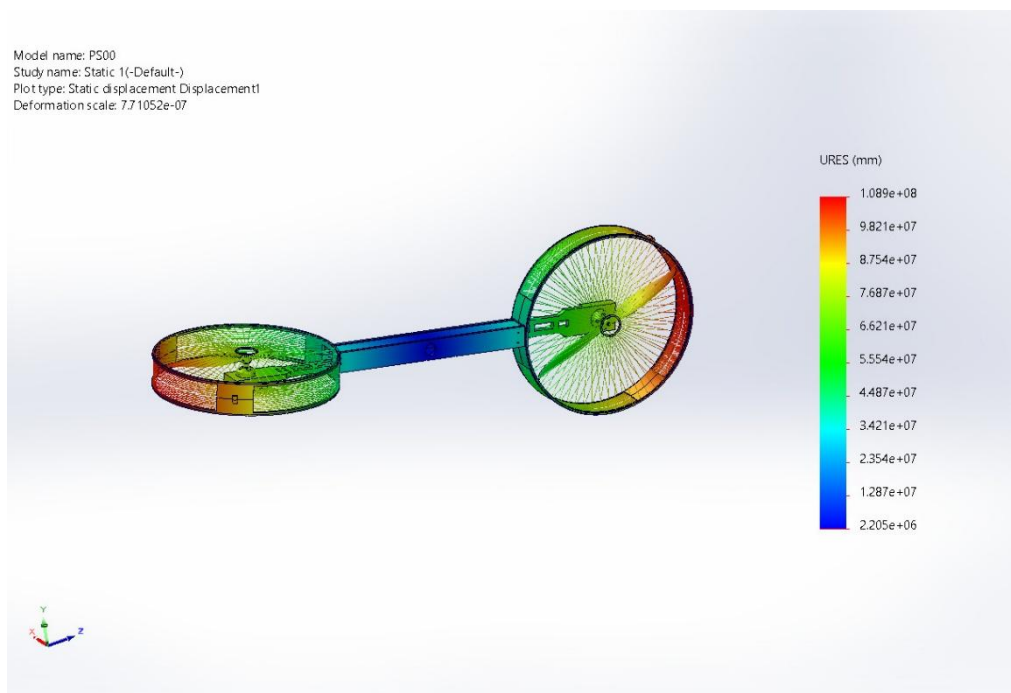


Figure 10 : Displacement on Pitch Structure

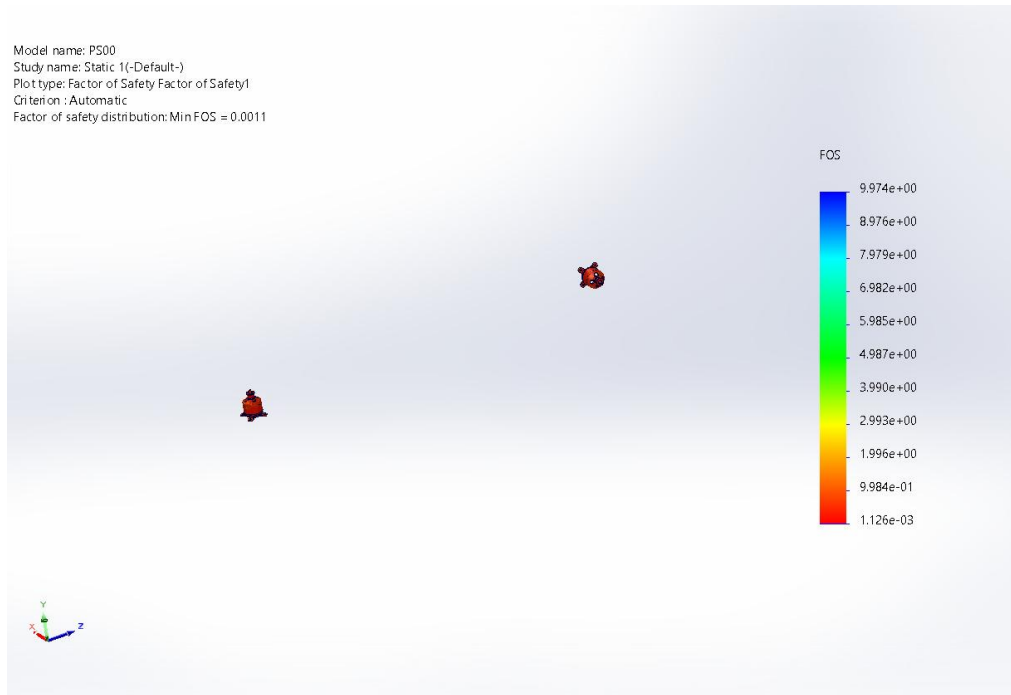


Figure 11 : Factor of Safety on Pitch structure

4.4. Torque Simulation of Yaw Structure

Yaw motion is caused by torque generated by difference of thrust between pitch motor and yaw motor.

For simulation, let's assume only yaw motor produce safe continues thrust,
So, ΔT is 7N.

Yaw torque acting on the yaw structure,

$$c = \Delta T \times r \quad \dots(38)$$

Let τ be torque and r be distance from motor to POR,

$$\tau = 7 \times 0.285$$

$$\tau = 1.995 \text{ Nm}$$

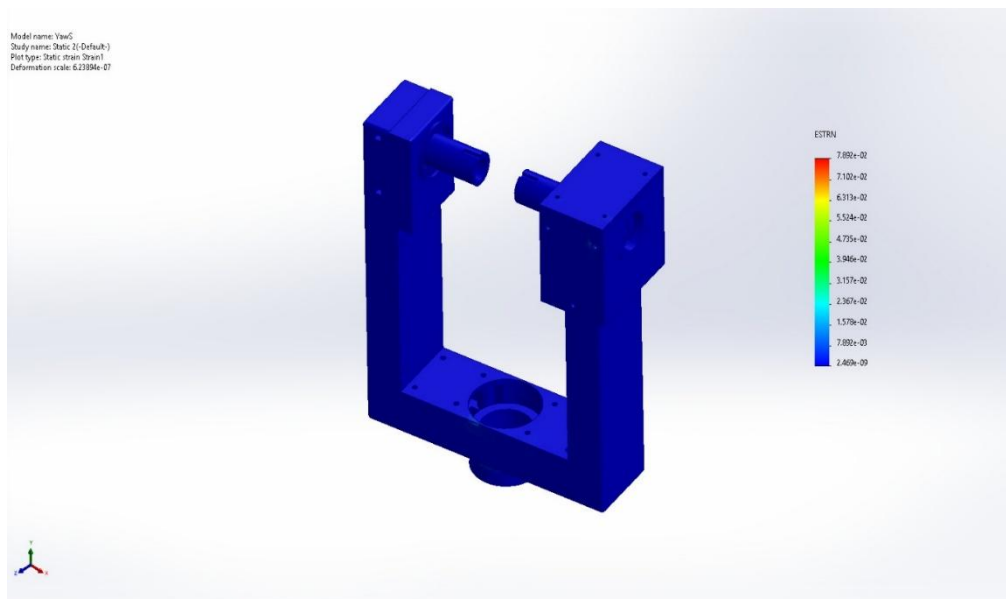


Figure 12 : Strain on Yaw Structure

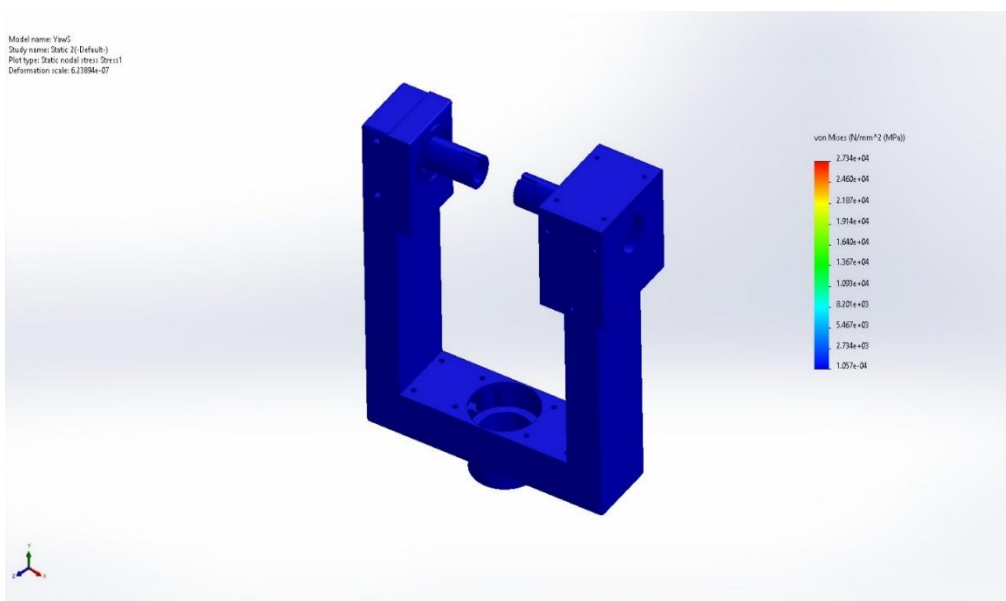


Figure 13 : Stress on Yaw Structure

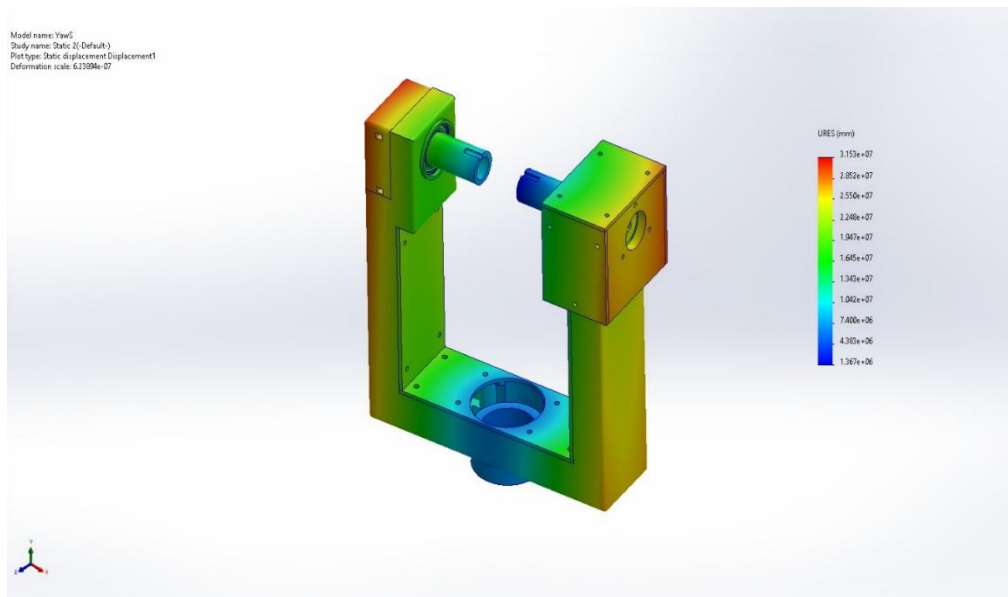


Figure 14 : Displacement of Yaw structure

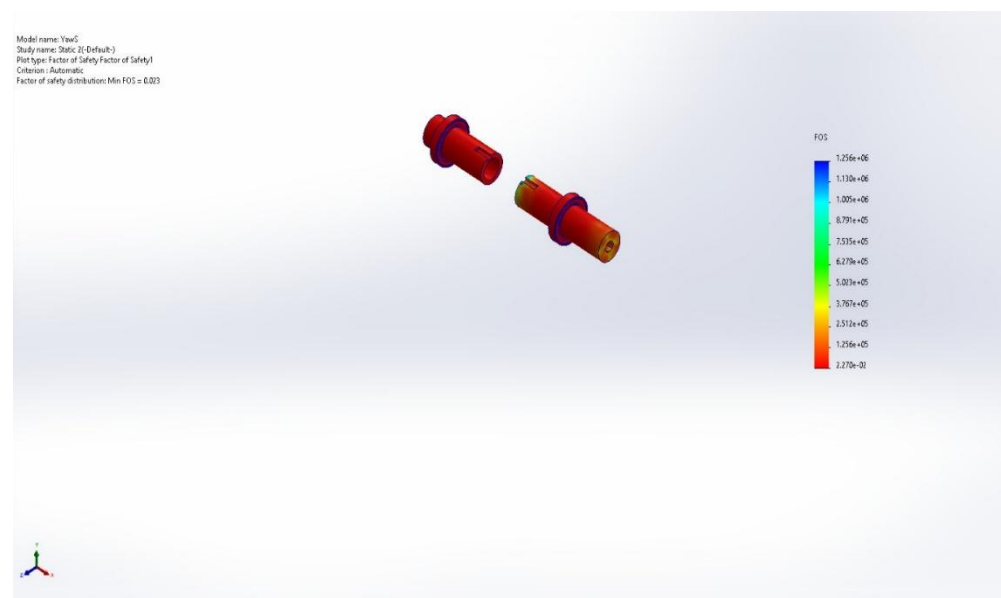


Figure 15 : Factor of Safety on Yaw Structure

4.5. Load Simulation on Vertical shaft and Bearing housing

The rotating structure is mounted on the vertical shaft there for, Load from weight of rotating structure carried by vertical shaft and distributes to bearing housing.

Total weight of the rotating structure is 863.37g.

$$F = mg = 0.86337kg \times 9.81 ms^{-2} = 8.47N$$

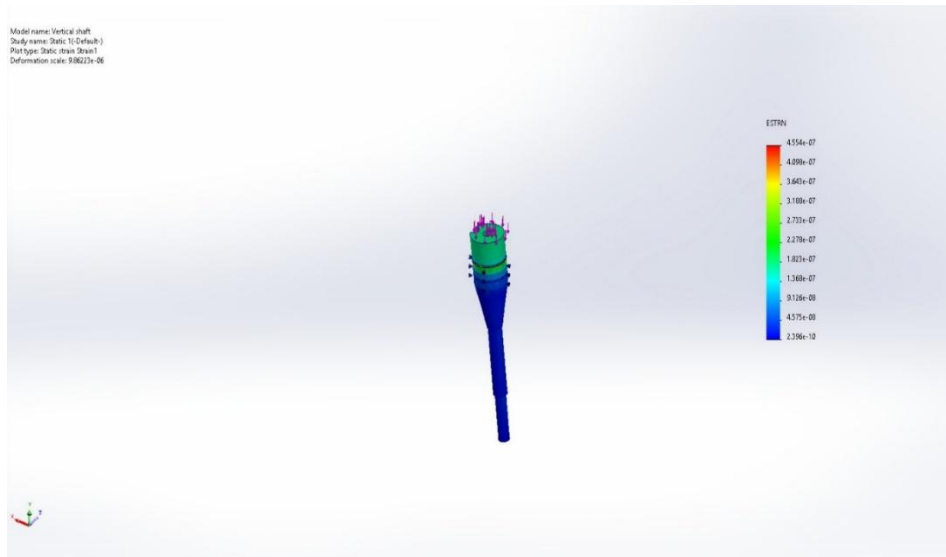


Figure 16 : Strain on Vertical Shaft

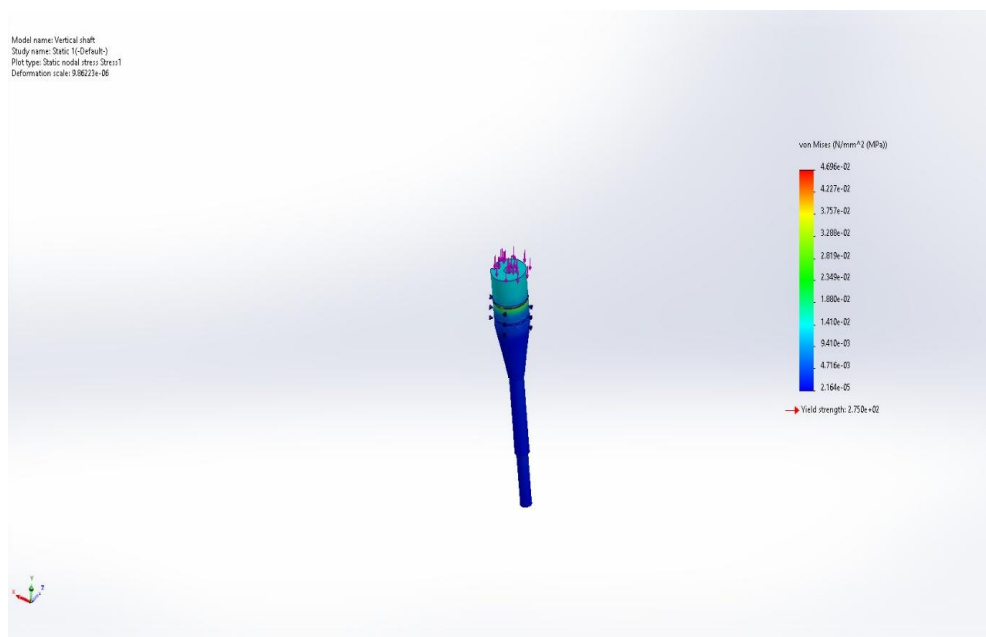


Figure 17 : Stress on Vertical Shaft

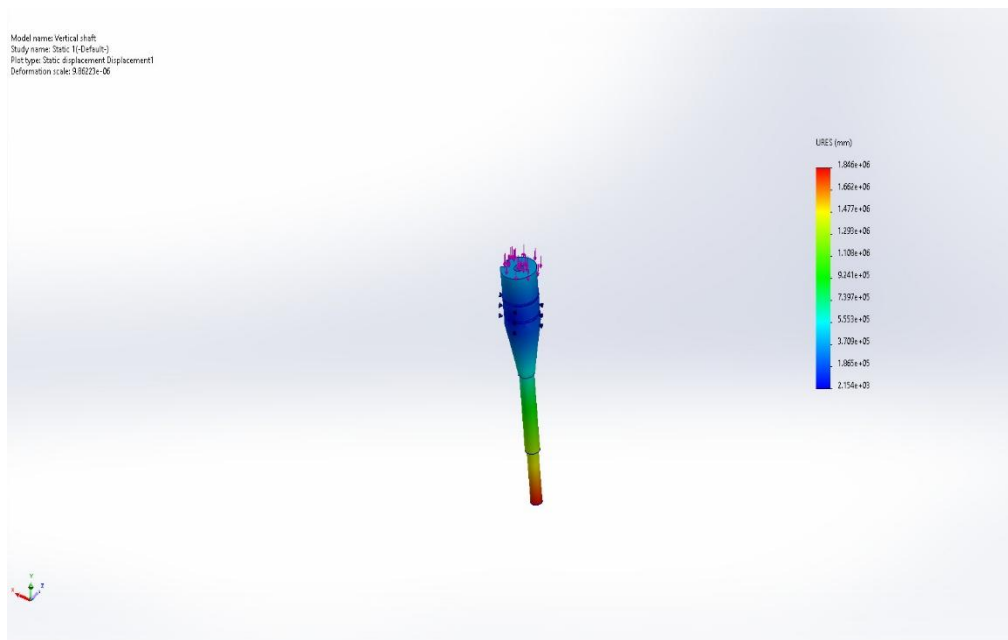


Figure 18 : Displacement on Vertical Shaft

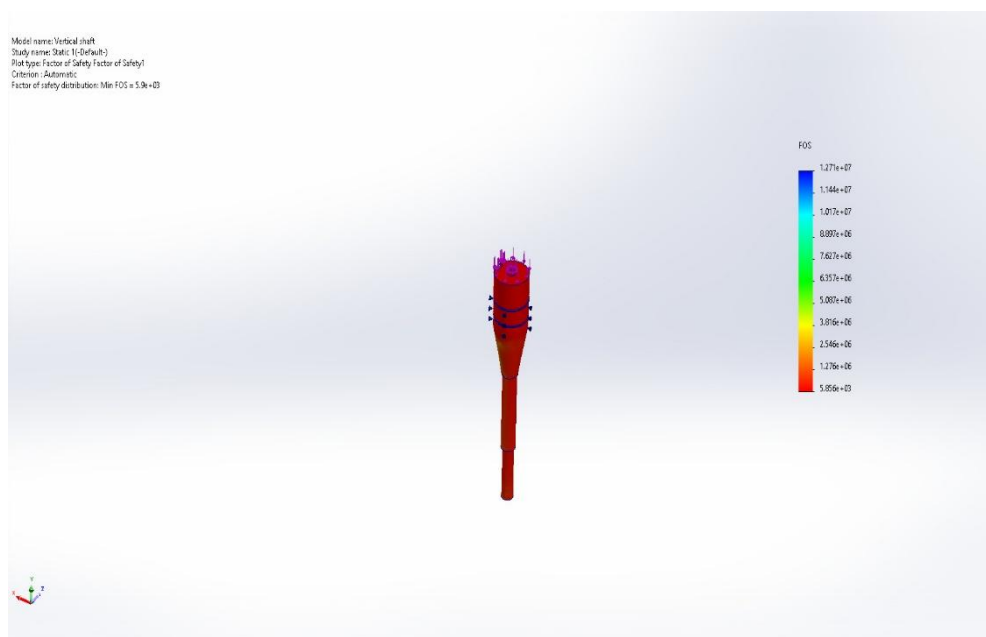


Figure 19 : Factor of Safety on Vertical Shaft

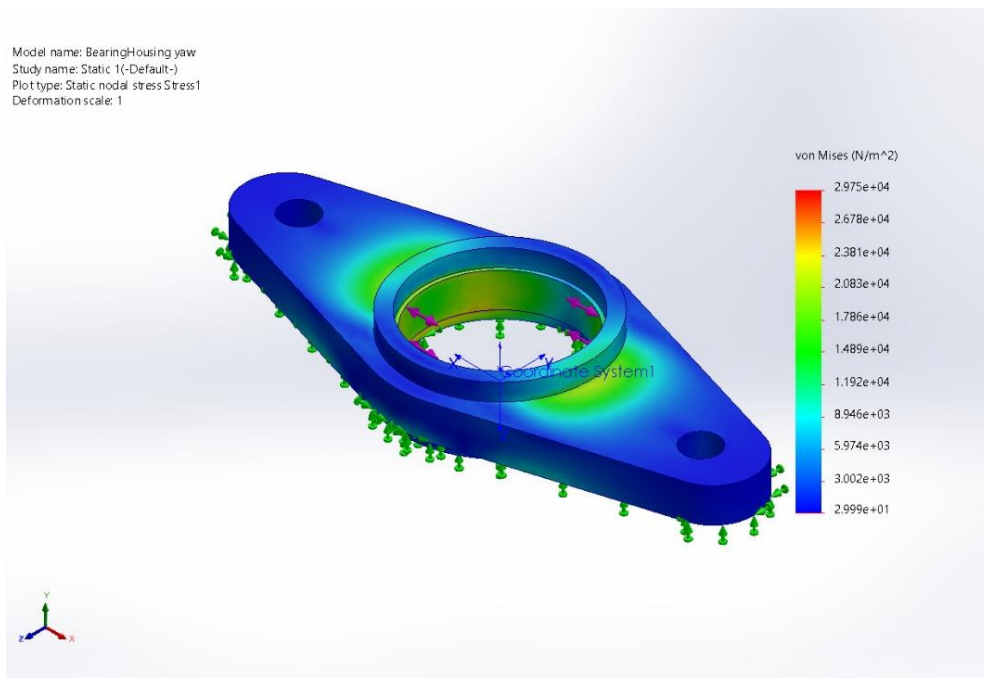


Figure 20 : Stress due to bearing load

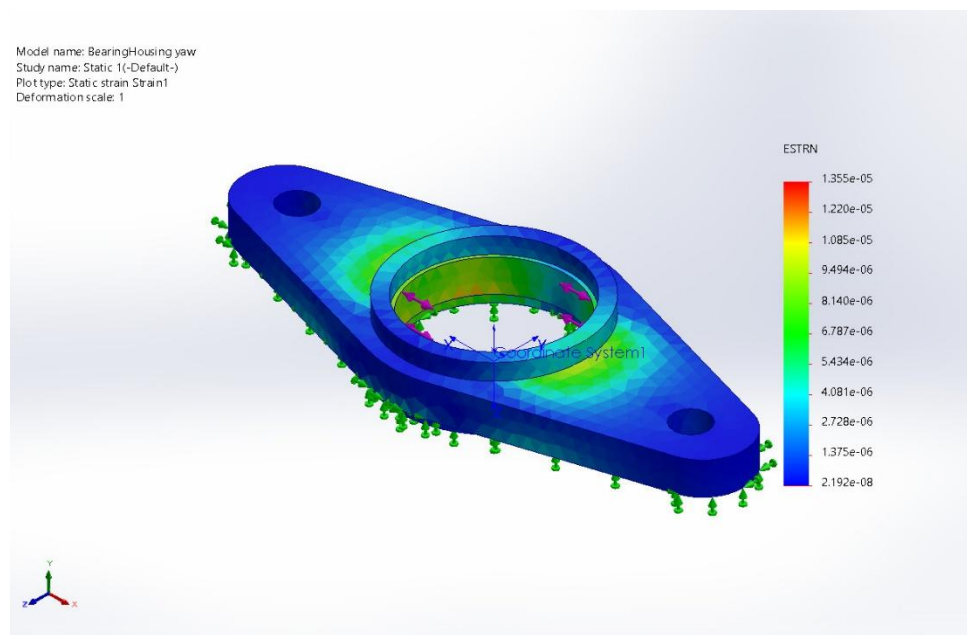


Figure 21 : Strain due to bearing load

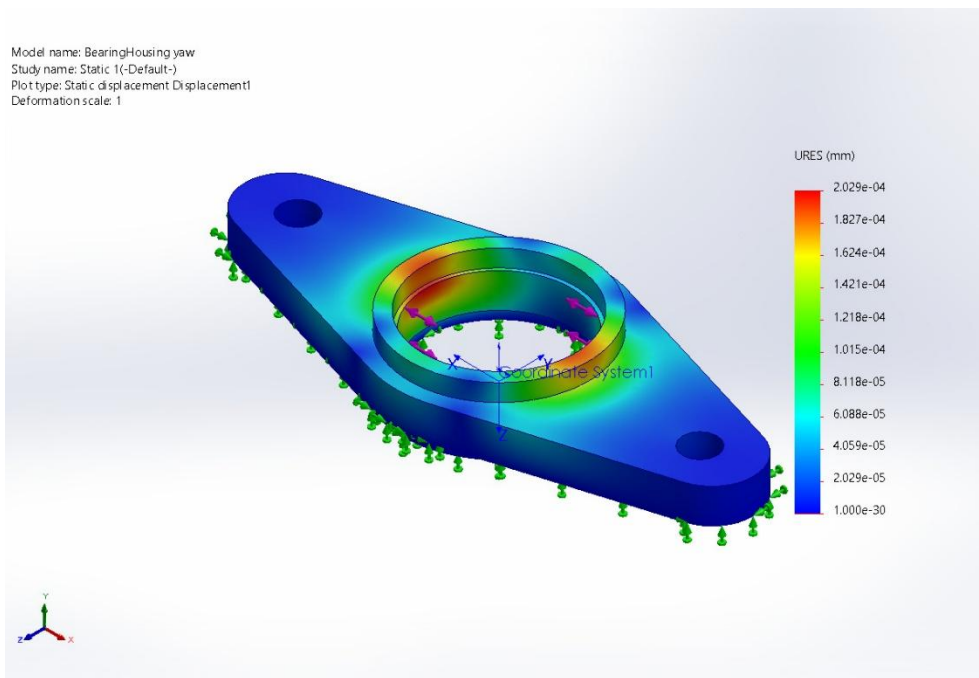


Figure 22 : Displacement due to bearing load

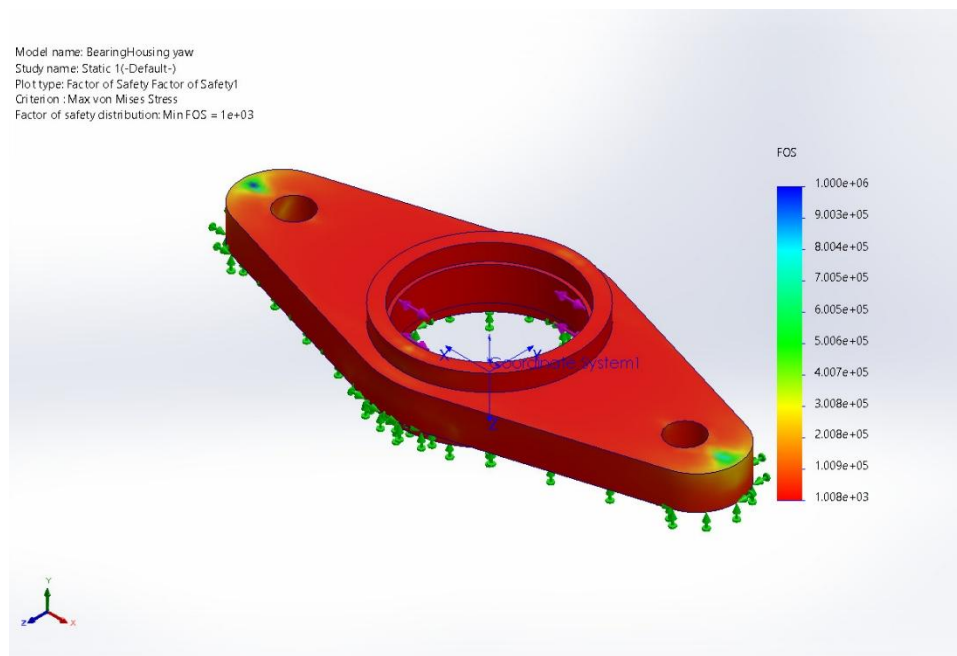


Figure 23 : Factor of safety due to bearing load

5. Conclusion and recommendations

At this stage of the project, the conceptual design, mechanical configuration, material selection, and preliminary mathematical modeling of the 2-DOF twin rotor helicopter test bed have been successfully completed. Structural calculations confirm that the selected materials and dimensions satisfy safety and fatigue requirements for laboratory operation, while the chosen motors and propulsion system provide adequate thrust and control authority.

The derived dynamic model forms a solid foundation for controller development, and the selection of PID control combined with an STM32-based real-time platform ensures feasibility for practical implementation. Future work will focus on controller tuning, MATLAB/Simulink validation, hardware integration, and experimental testing to verify system performance against theoretical predictions.

Overall, the project is progressing as planned and is well-positioned to achieve its objective of developing a reliable and educational 2-DOF helicopter control test bed for advanced control system studies.

6. Acknowledgment

We would like to express our sincere gratitude to all those who have provided guidance and support during Project Progress 1 of our final year group project.

First and foremost, we would like to extend our sincere appreciation to our project supervisor, Eng. H. G. C. R. Laksiri, for his valuable guidance, continuous supervision, and constructive feedback provided during project. His technical knowledge, encouragement, and advice have been instrumental in helping us to understand the project requirements and to proceed in the correct direction.

We also wish to express our sincere thanks to the Final Year Project Coordinator, Dr. Eng. J. M. N. Jayaweera, for her guidance, coordination, and support throughout the project progress evaluation process.

Furthermore, we would like to acknowledge the academic and non-academic staff of the Department of Engineering Technology, Uva Wellassa University of Sri Lanka, for providing the necessary facilities, resources, and support required for the successful completion of this project progress.

Finally, we would like to express our appreciation to our families and peers for their continuous encouragement, patience, and moral support throughout the course of our undergraduate studies and during the early stages of this project.

7. References

- [1] P. Chalupa, J. Přikryl, and J. Novák, "Modelling of twin rotor MIMO system," in *Procedia Engineering*, Elsevier Ltd, 2015, pp. 249–258. doi: 10.1016/j.proeng.2015.01.365.
- [2] M. Reyhanoglu, M. Jafari, and M. Rehan, "Simple Learning-Based Robust Trajectory Tracking Control of a 2-DOF Helicopter System," *Electronics (Switzerland)*, vol. 11, no. 13, Jul. 2022, doi: 10.3390/electronics11132075.
- [3] "Twin Rotor MIMO 33-007-PCI." [Online]. Available: www.feedback-instruments.com
- [4] "ahma_02".
- [5] A. T. Azar, A. S. Sayed, A. S. Shahin, H. A. Elkholy, and H. H. Ammar, "PID Controller for 2-DOFs Twin Rotor MIMO System Tuned with Particle Swarm Optimization," *Advances in Intelligent Systems and Computing*, vol. 1058, pp. 229–242, 2020, doi: 10.1007/978-3-030-31129-2_22.
- [6] J. Sivadasan and J. R. J. Shiney, "Modified nondominated sorting genetic algorithm-based multiobjective optimization of a cross-coupled nonlinear PID controller for a Twin Rotor System," *Journal of Engineering and Applied Science*, vol. 70, no. 1, Dec. 2023, doi: 10.1186/S44147-023-00305-6.

8. Appendices

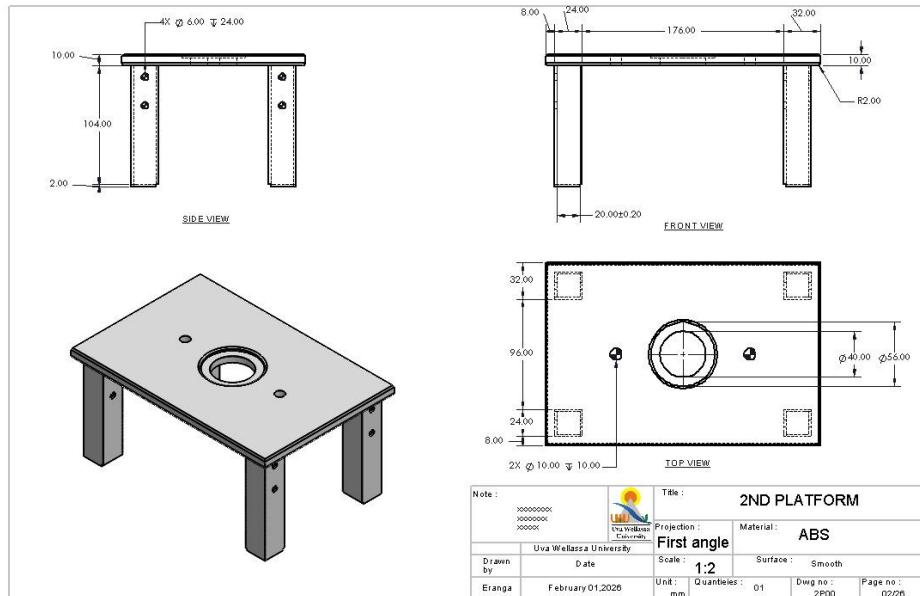


Figure 25 : 2D Drawing of 2nd Platform

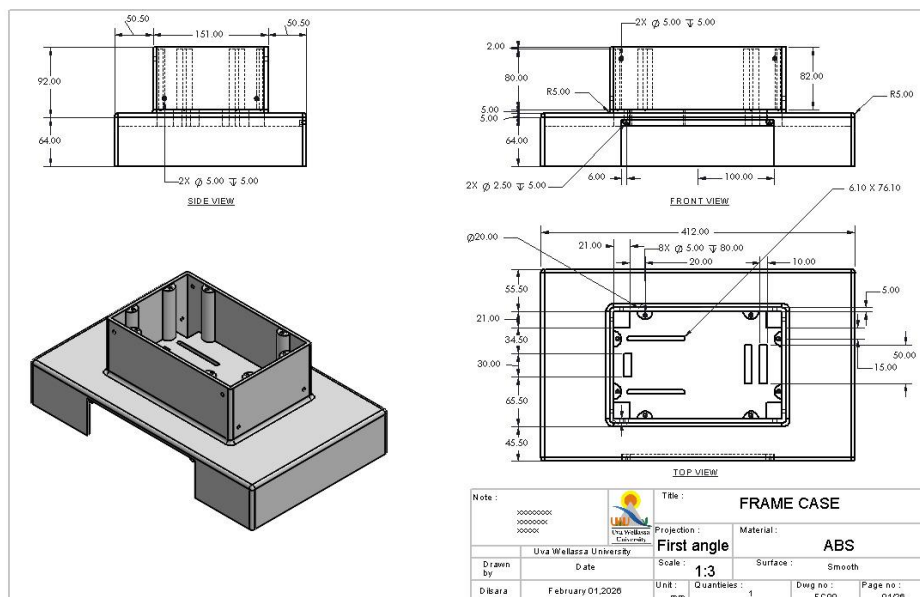


Figure 24 : 2D Drawing of Support Frame Case

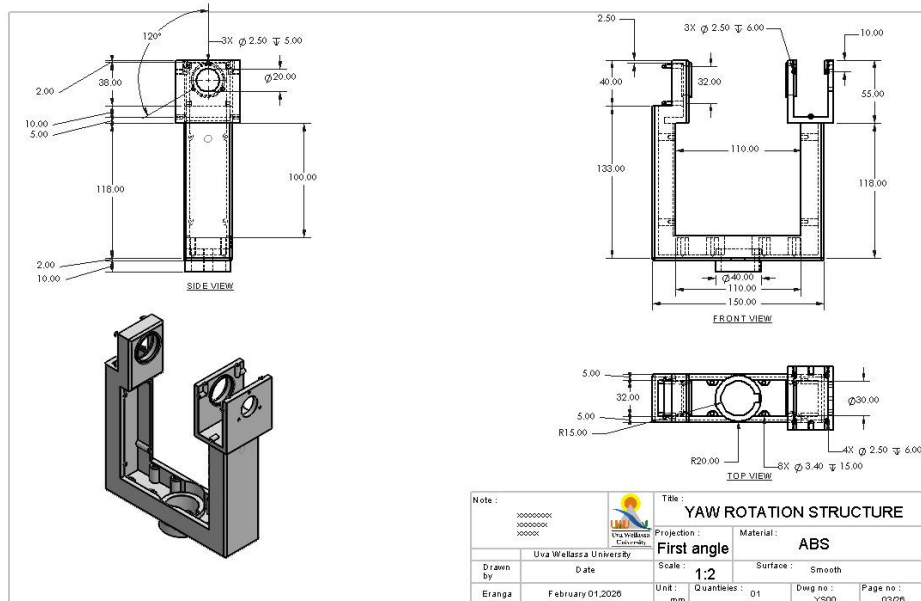


Figure 26 : 2D Drawing of Yaw Rotation

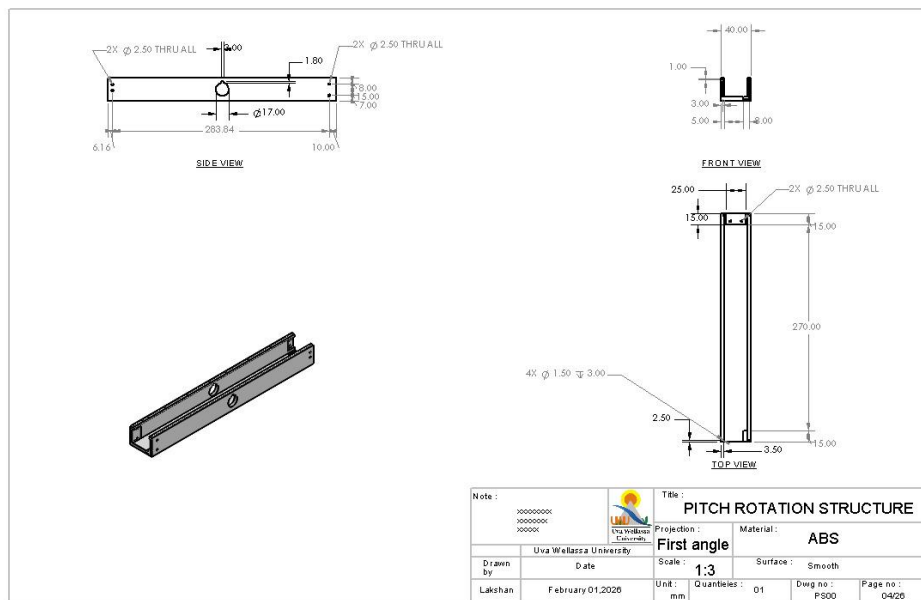


Figure 27 : 2D Drawing of Pitch Rotation Structure

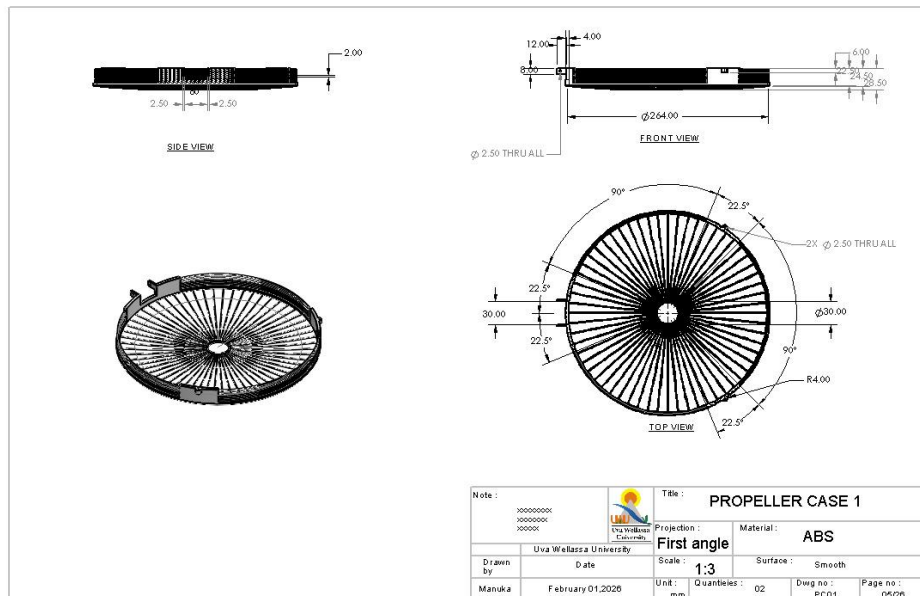


Figure 28 : 2D Drawing of Propeller Case 1

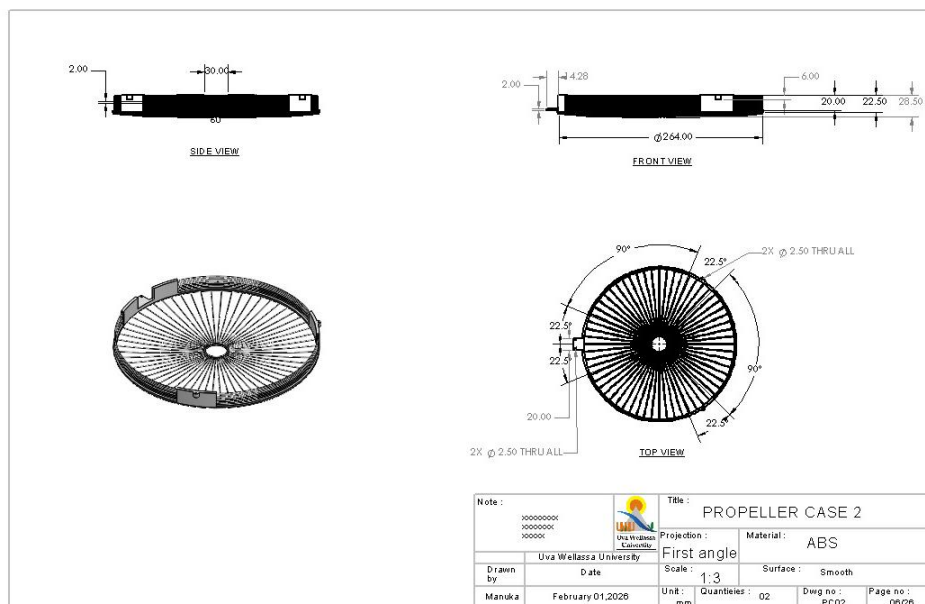


Figure 29 : 2D Drawing of Propeller Case 2

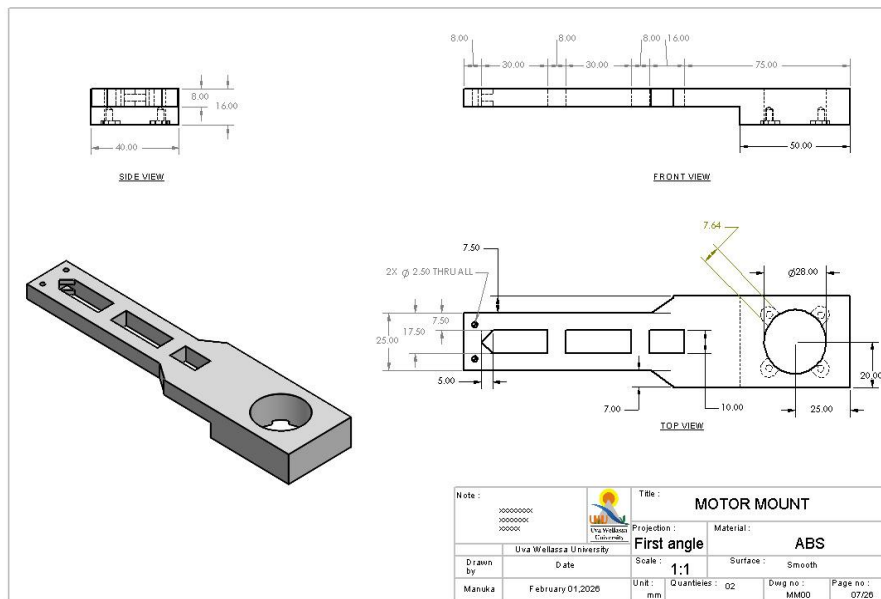


Figure 30 : 2D Drawing of Motor Mount

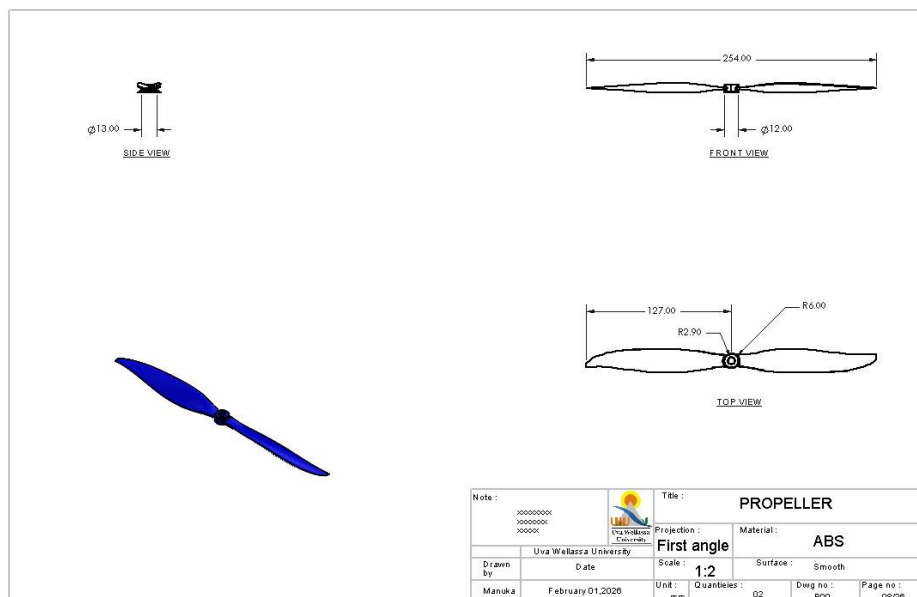


Figure 31 : 2D Drawing of Propeller

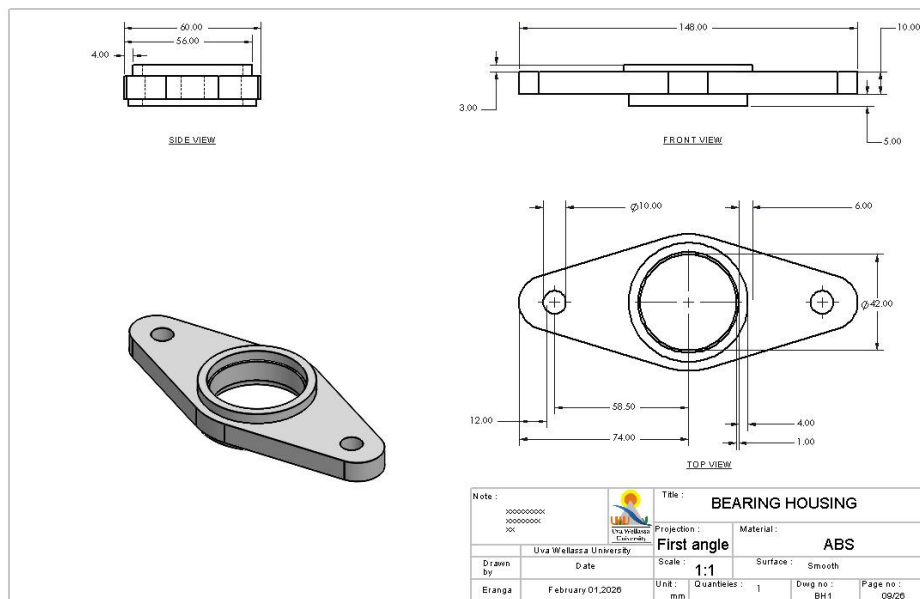


Figure 32 : 2D Drawing of Bearing Housing

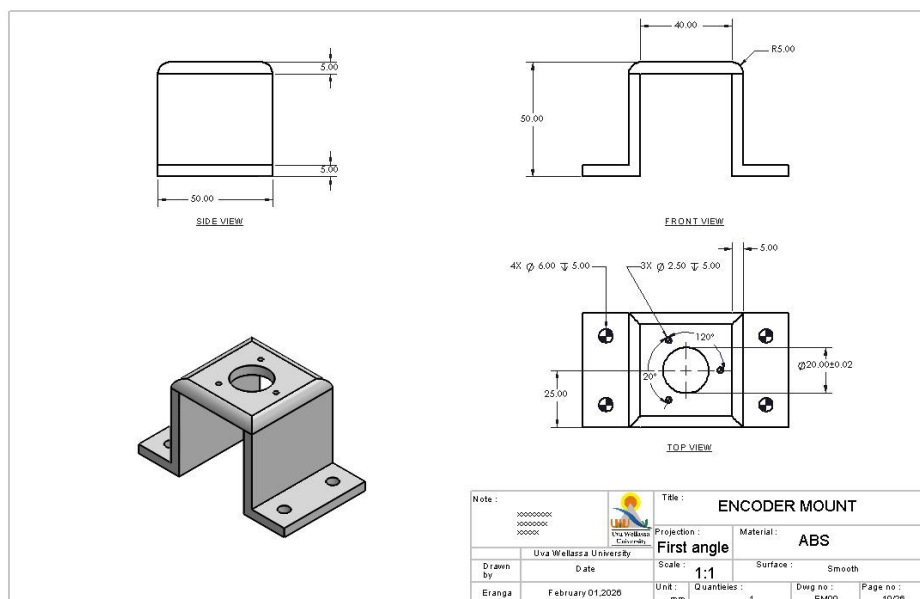


Figure 33 : 2D Drawing of Encoder Mount

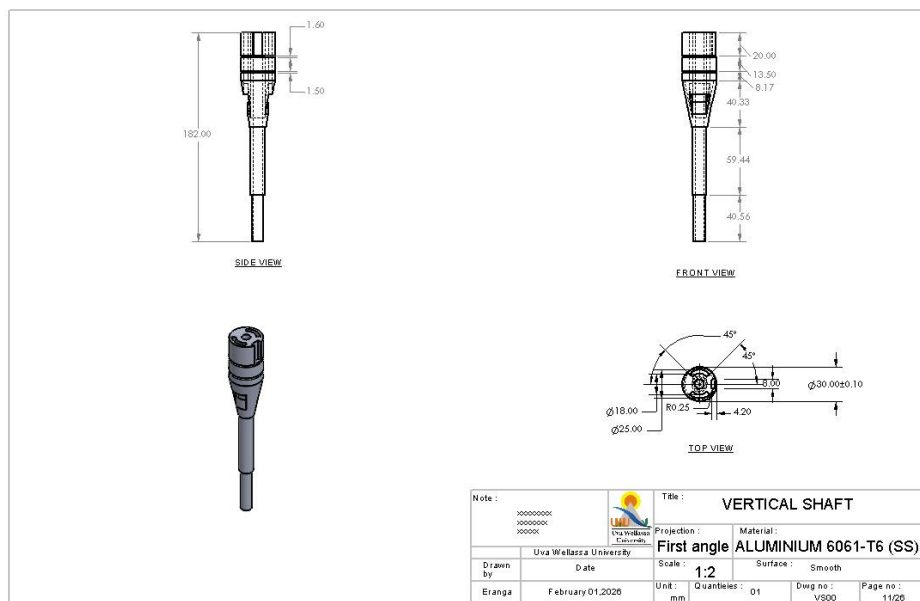


Figure 34 : 2D Drawing of Vertical Shaft

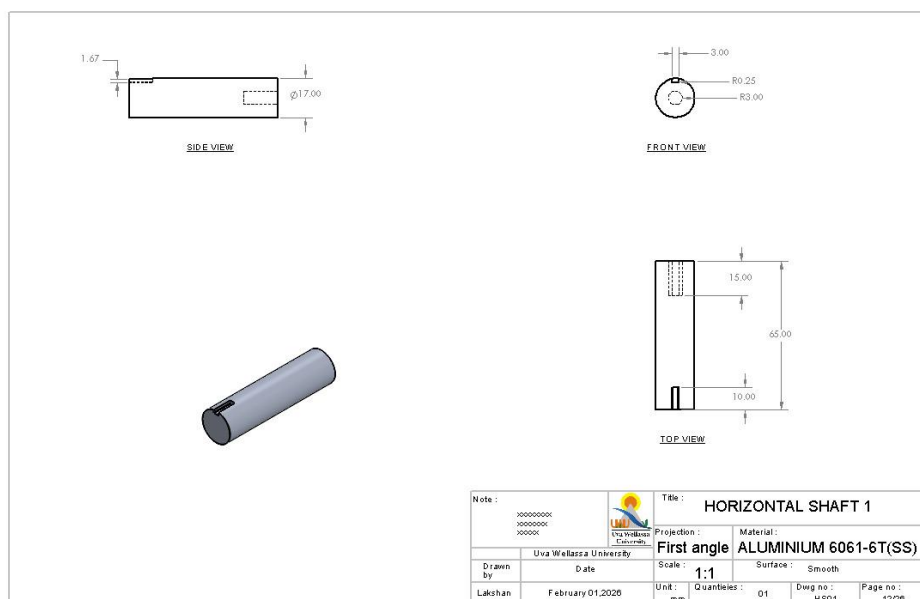


Figure 35 : 2D Drawing of Horizontal Shaft 1

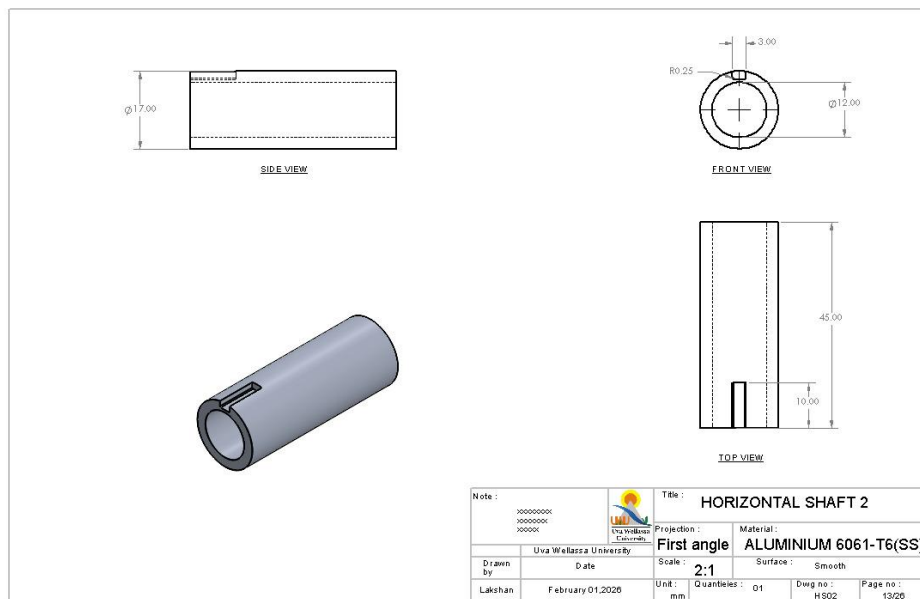


Figure 36 : 2D Drawing of Horizontal Shaft 2

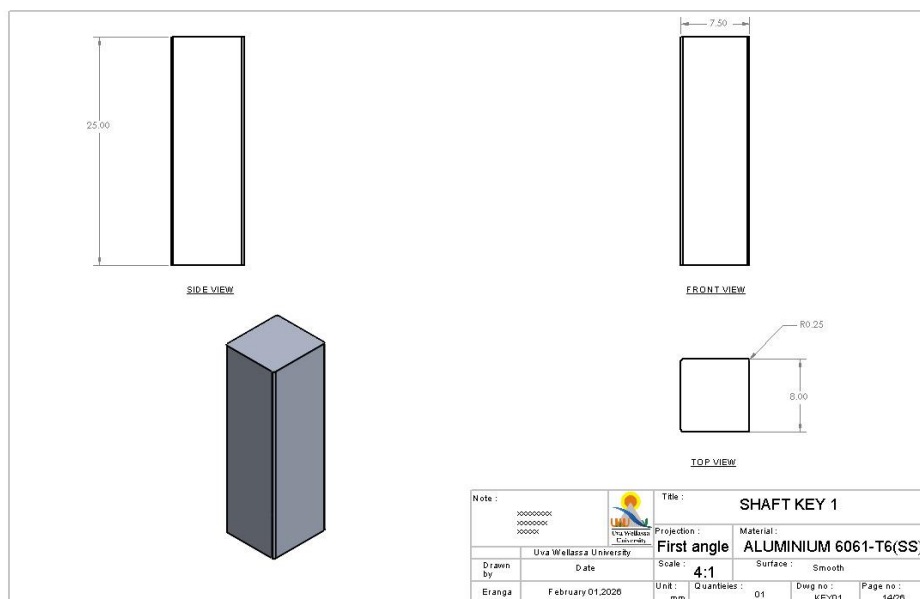


Figure 37 : 2D Drawing of Shaft Key 1

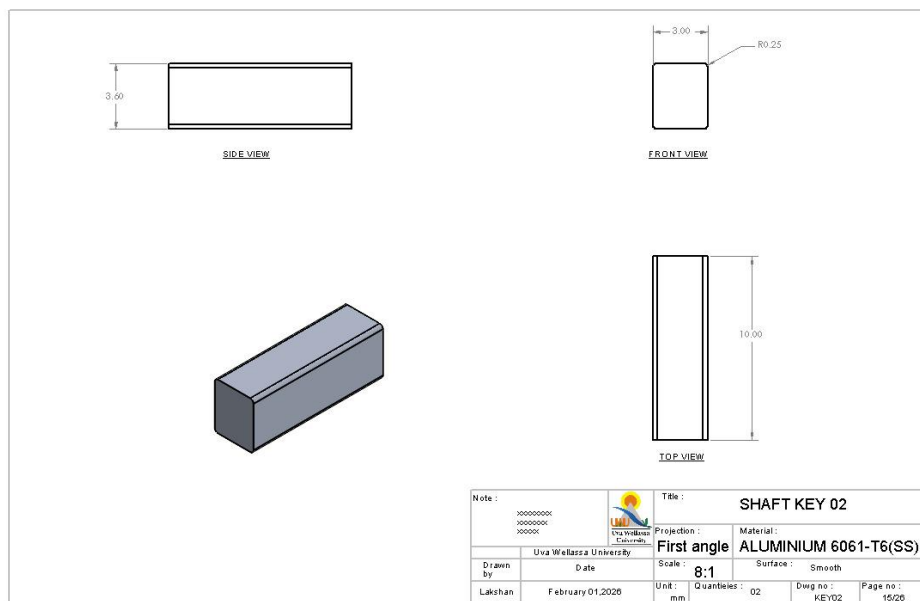


Figure 38 : 2D Drawing of Shaft Key 2

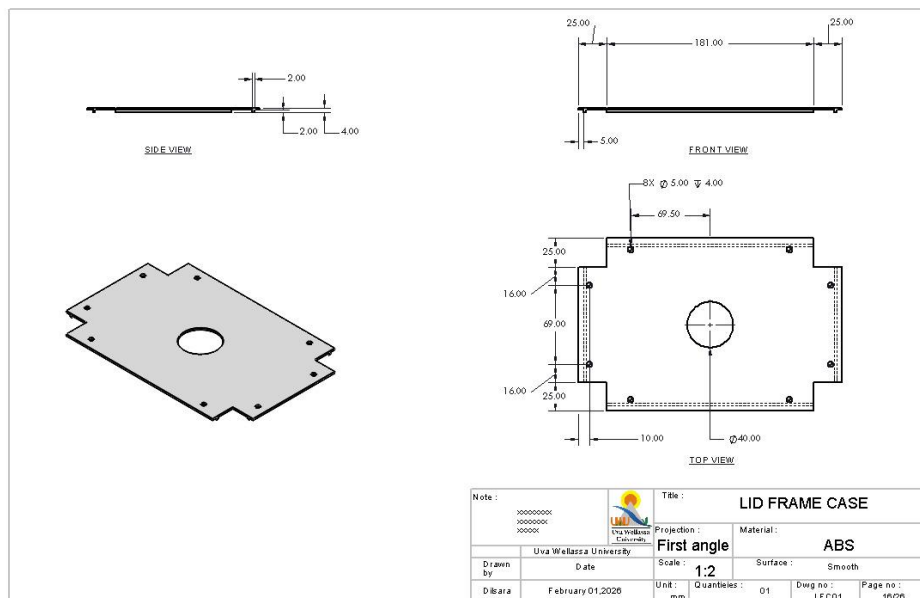


Figure 39 : 2D Drawing of LID Frame Case

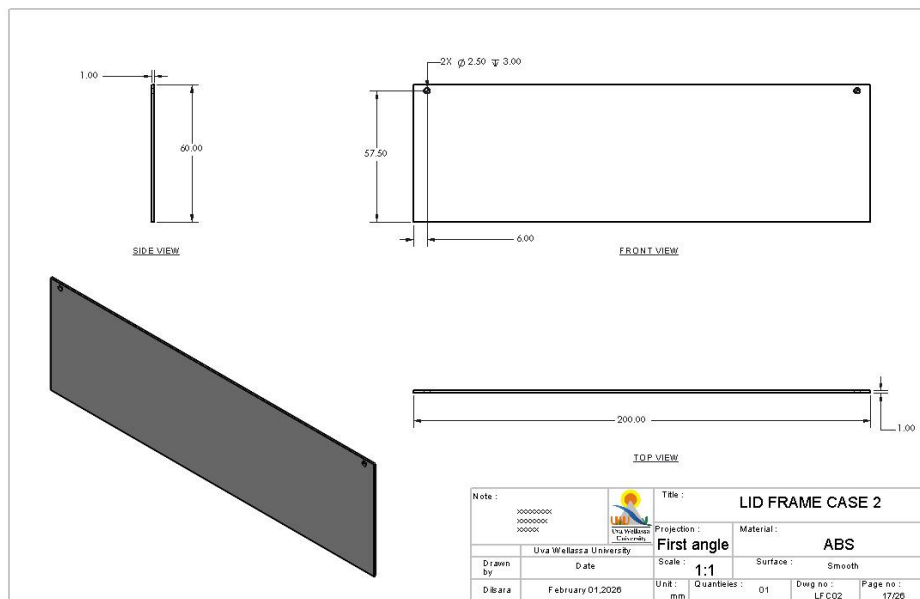


Figure 40 : 2D Drawing of LID Frame Case 2

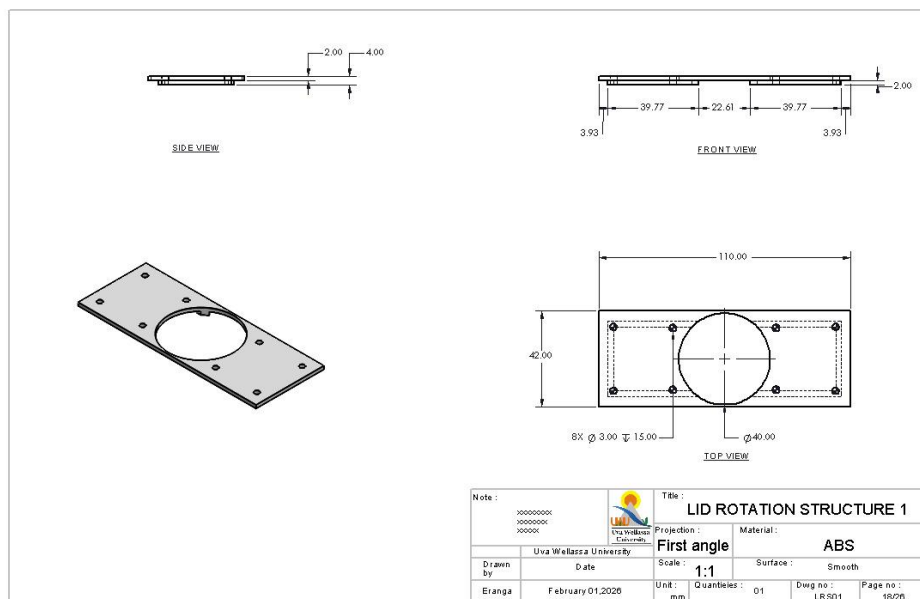


Figure 41 : 2D Drawing of LID Rotation Structure 1

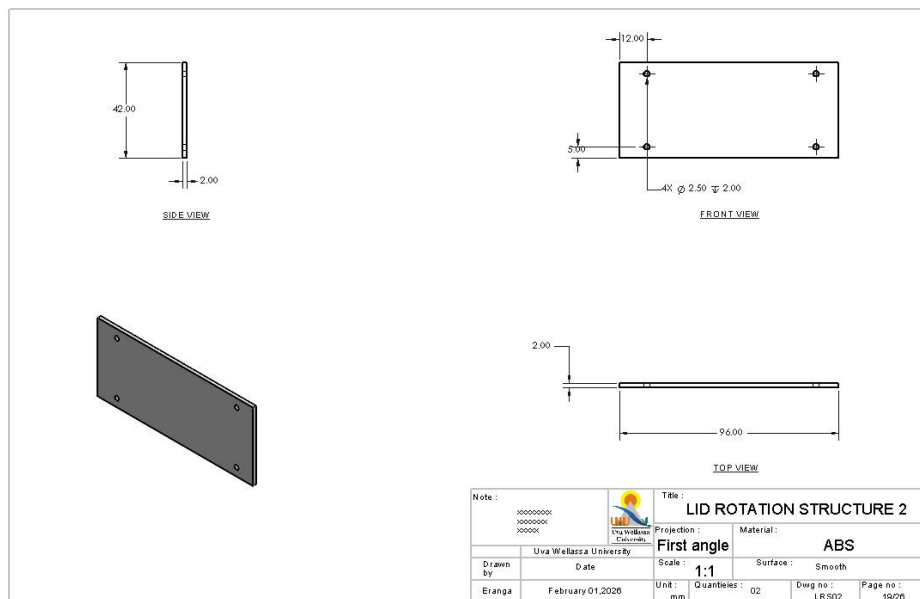


Figure 42 : 2D Drawing of LID Rotation Structure 2

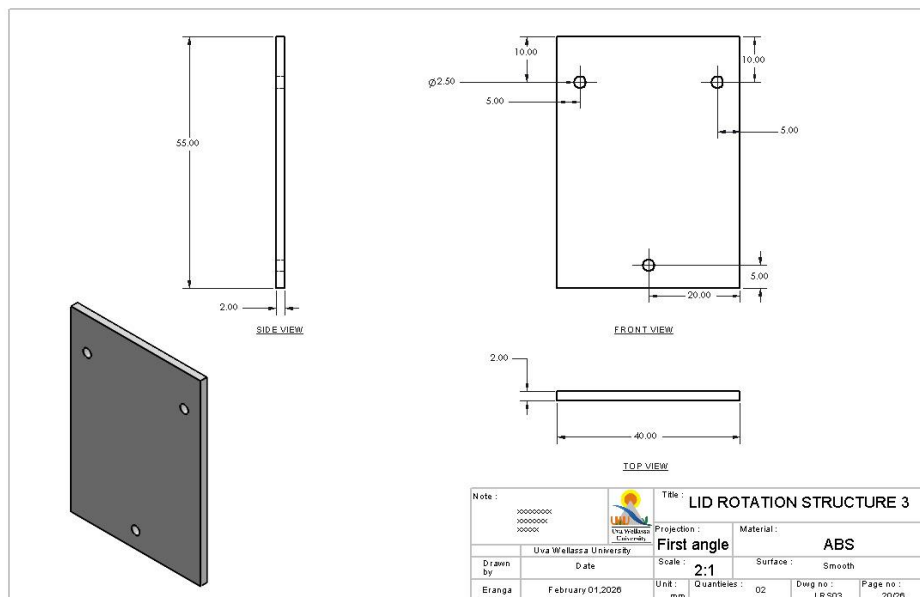


Figure 43 : 2D Drawing of LID Rotation Structure 3

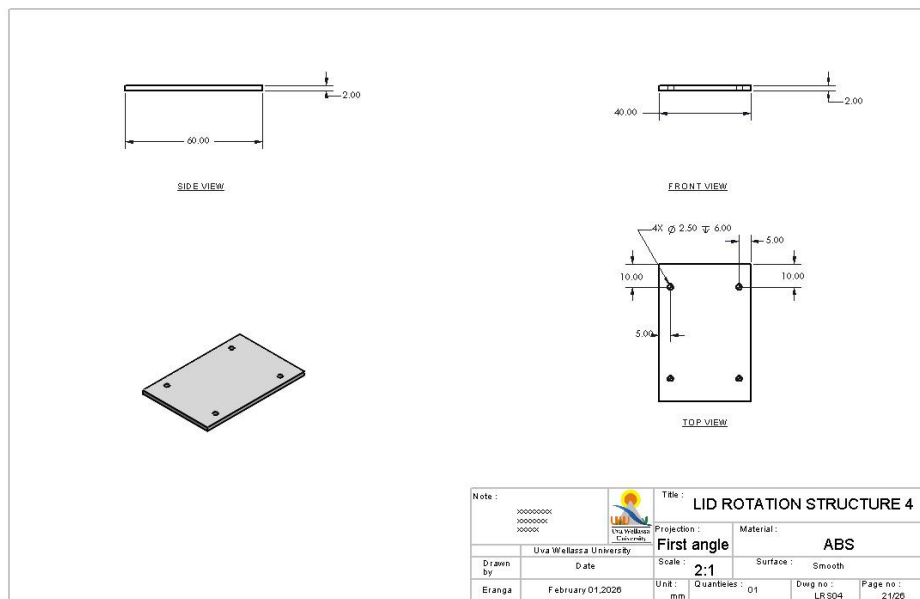


Figure 44 : 2D Drawing of LID Rotation Structure 4

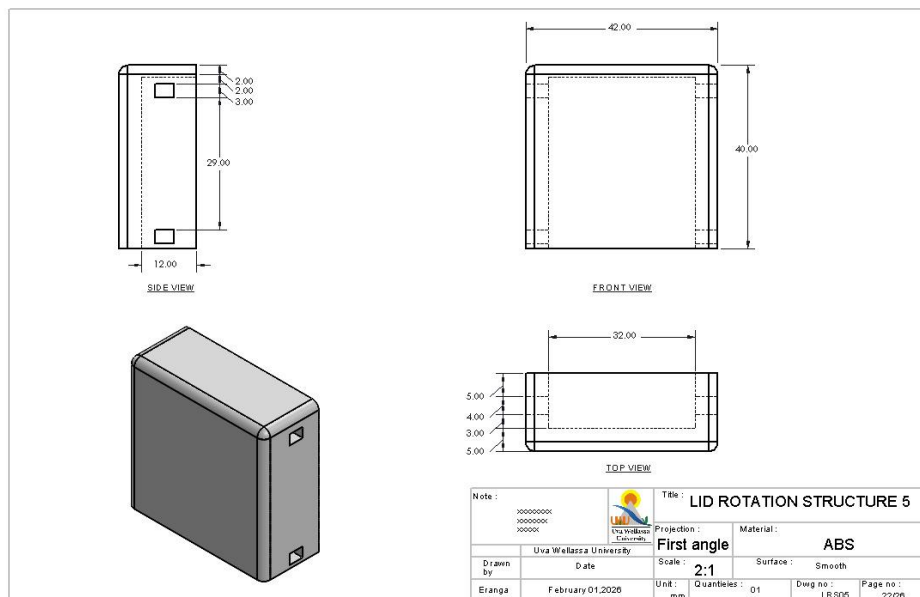


Figure 45 : 2D Drawing of LID Rotation Structure 5

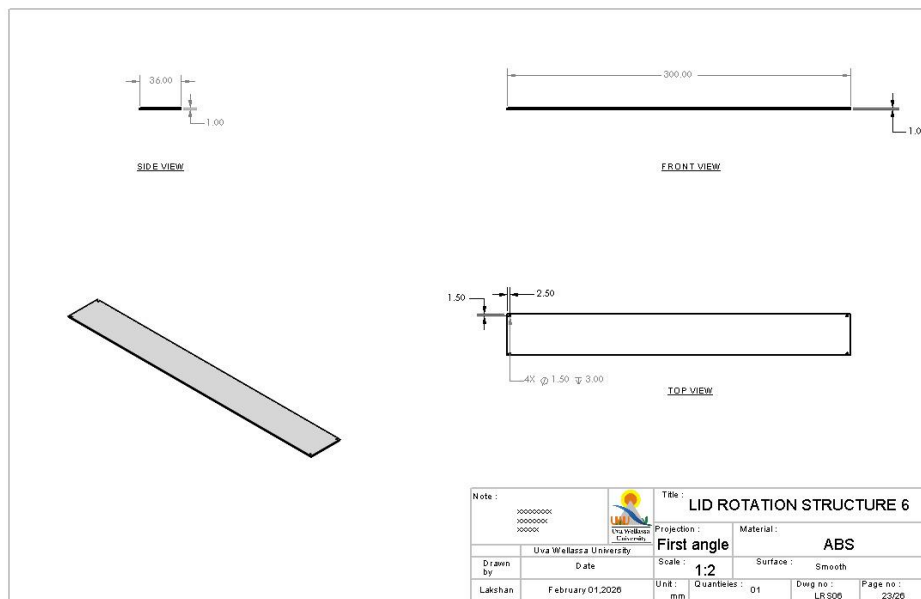


Figure 46 : 2D Drawing of LID Rotation Structure 6



Figure 47 : 2D Drawing of Wood Base

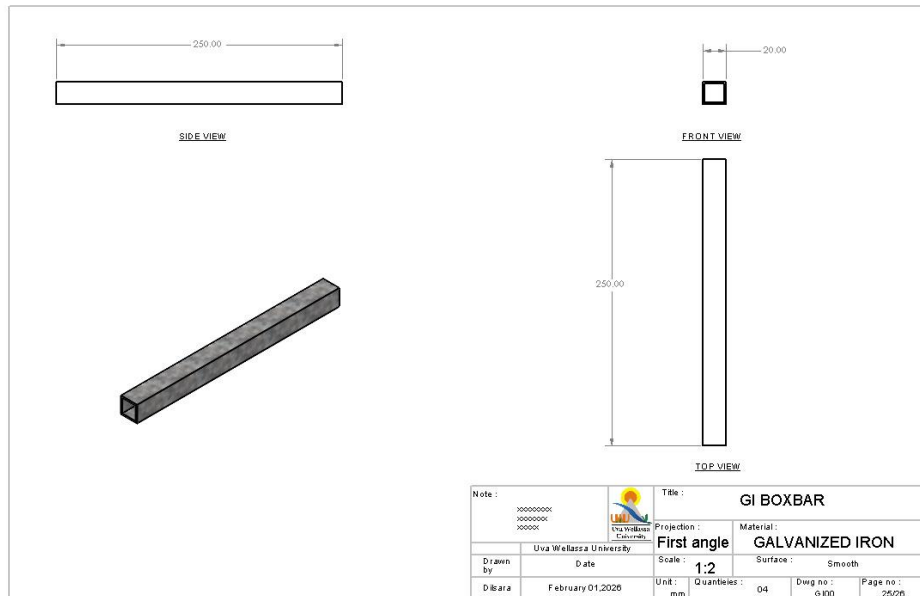


Figure 48 : 2D Drawing of GI Box Bar

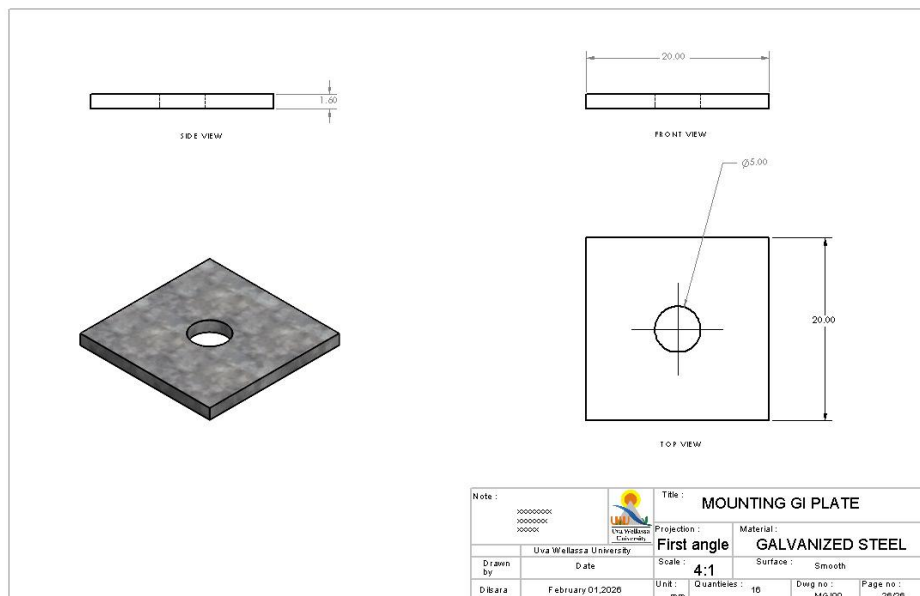


Figure 49 : 2D Drawing of Mounting GI Plate